

DAYANANDA SAGAR UNIVERSITY

Devarakaggalahalli, Harohalli, Kanakapura Road, Bengaluru South Dt. – 562 112

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in
COMPUTER SCIENCE AND ENGINEERING**

Major Project Phase-II Report

AI-ASSISTED TB DIAGNOSIS AND DRUG RESISTANCE PREDICTION IN HIGH- BURDEN DISTRICTS OF KARNATAKA

Batch: 59

By

**Sampreeth S M – ENG22CS0437
Samriddhi Panigrahi – ENG22CS0438
Satvik D Joshi - ENG22CS0446
Vaishnavi Handa - ENG22CS0493**

**Under the supervision of
Dr. Kumar Dilip
Associate Professor
Department of Computer Science and Engineering**

**DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
SCHOOL OF ENGINEERING
DAYANANDA SAGAR UNIVERSITY**

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Dayananda Sagar University
Devarakaggalahalli, Harohalli, Kanakapura Road, Bengaluru South Dt. – 562 112
Department of Computer Science & Engineering

CERTIFICATE

This is to certify that the Phase-II project work titled “**AI-ASSISTED TB DIAGNOSIS AND DRUG RESISTANCE PREDICTION IN HIGH- BURDEN DISTRICTS OF KARNATAKA**” is carried out by **Sampreeth S M (ENG22CS0437), Samriddhi Panigrahi (ENG22CS0438), Satvik D Joshi (ENG22CS0446), Vaishnavi Handa (ENG22CS0493)**, bonafide students of Bachelor of Technology in Computer Science and Engineering at the School of Engineering, Dayananda Sagar University, Bangalore in partial fulfillment for the award of degree in Bachelor of Technology in Computer Science and Engineering, during the year **2024-2025**.

Dr. Kumar Dilip

Associate Professor
Dept. of CS&E,
School of Engineering
Dayananda Sagar University

Date:

Dr. Girisha G S

Chairman CSE
School of Engineering
Dayananda Sagar University

Date:

**Dr. Udaya Kumar Reddy
K R**

Dean
School of Engineering
Dayananda Sagar University

Date:

Name of the Examiner

Signature of Examiner

1.

2.

DECLARATION

We, **Sampreeth S M (ENG22CS0437), Samriddhi Panigrahi (ENG22CS0446), Satvik D Joshi (ENG22CS0446), Vaishnavi Handa (ENG22CS0493)**, are students of eighth semester B. Tech in **Computer Science and Engineering**, at School of Engineering, **Dayananda Sagar University**, hereby declare that the Major Project Stage-II titled “**AI-Assisted TB Diagnosis and Drug Resistance Prediction in High- Burden Districts of Karnataka**” has been carried out by us and submitted in partial fulfilment for the award of degree in **Bachelor of Technology in Computer Science and Engineering** during the academic year **2025-2026**.

Student

Signature

Name 1: Sampreeth S M

USN: ENG22CS0437

Name 2: Samriddhi Panigrahi

USN: ENG22CS0438

Name 3: Satvik D Joshi

USN: ENG22CS0446

Name 4: Vaishnavi Handa

USN: ENG22CS0493

Place: Bangalore

Date:

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NOMENCLATURE USED

Term	Definition
AUC-ROC	Area Under the Receiver Operating Characteristic Curve
CBAM	Convolutional Block Attention Module
CLAHE	Contrast Limited Adaptive Histogram Equalization
DR-TB	Drug-Resistant Tuberculosis
FN	False Negative
FP	False Positive
Grad-CAM	Gradient-weighted Class Activation Mapping
LIME	Local Interpretable Model-agnostic Explanations
PACS	Picture Archiving and Communication System
RF	Random Forest
SHAP	SHapley Additive exPlanations
TB	Tuberculosis
TN	True Negative
TP	True Positive
XAI	Explainable Artificial Intelligence

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ABSTRACT

This research presents the development and evaluation of an AI-assisted diagnostic framework for tuberculosis (TB) detection and drug-resistance prediction, specifically designed for the resource-constrained healthcare infrastructure of Karnataka, India. The system implements a two-branch multimodal architecture: an imaging branch utilizing EfficientNet-V2 with Convolutional Block Attention Modules (CBAM) for automated chest radiograph analysis, and a tabular branch employing an XGBoost-Random Forest ensemble for predicting drug resistance from genomic mutations and patient metadata.

Experimental results on a geographically held-out test set demonstrate that the imaging branch achieves a state-of-the-art AUC-ROC of 0.950 for TB screening. Furthermore, the multimodal fusion approach significantly outperforms single-modality baselines, yielding a drug-resistance prediction AUC of 0.961—representing an 18.8-point improvement over image-only models and a 5.7-point gain over genomic-only approaches. To ensure clinical adoption, the system integrates triple-layer explainability using Grad-CAM++, LIME, and SHAP, providing clinicians with interpretable visual heatmaps and feature-level importance rankings. The framework is deployed via the NextTB web application, offering a scalable solution for early TB detection and comprehensive risk assessment in district-level health centres.

Keywords: Deep Learning, Multimodal Data Fusion, Tuberculosis, Drug Resistance, Explainable AI (XAI), Medical Imaging.

CHAPTER 1

INTRODUCTION

CHAPTER 1 INTRODUCTION

1.1. BACKGROUND

Tuberculosis (TB) is an infectious disease caused by the bacterium *Mycobacterium tuberculosis*. Although it primarily targets the lungs, TB can affect virtually every organ system in the human body, including the kidneys, spine, and brain. It spreads through the air when an infected person coughs, sneezes, or speaks, making it exceptionally difficult to contain in densely populated settings.

Despite more than a century of scientific research, the development of the BCG vaccine, and the availability of effective antibiotic regimens since the 1950s, TB continues to be one of the deadliest infectious diseases in the world. According to the World Health Organization's 2023 Global Tuberculosis Report, approximately 10.6 million people fell ill with TB in 2022, and 1.3 million died [1]. This places TB among the top ten causes of death from a single infectious agent worldwide.

India bears a disproportionately heavy share of this global burden. The WHO estimates that India accounts for roughly 27% of all new TB cases annually, with over 2.8 million new cases notified in 2022 alone [1, 2]. The Government of India launched the National Tuberculosis Elimination Programme (NTEP) with an ambitious target of eliminating TB by 2025, five years ahead of the global Sustainable Development Goal target. Despite significant progress in case notification and treatment success rates, the country continues to grapple with deep structural challenges: fragmented healthcare delivery, stigma, diagnostic delays, and a growing burden of drug-resistant strains.

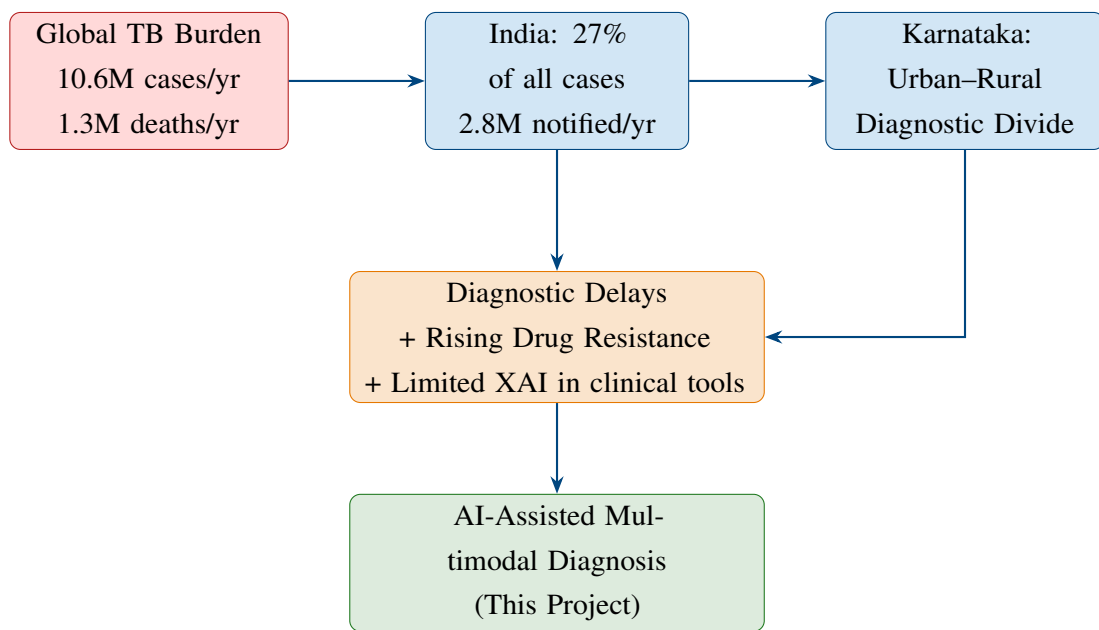


Figure 1.1: Motivational context: from global TB burden to AI-driven solution

Within India, the state of Karnataka presents a particularly instructive case study. The state hosts Bengaluru, one of India’s largest cities, where tertiary hospitals operate advanced digital radiography suites, GeneXpert MTB/RIF machines, and line-probe assay platforms around the clock. Yet, just a few hours’ drive away, in the northern and northeastern districts, many taluk-level primary health centres still depend almost exclusively on sputum smear microscopy for TB diagnosis, a technique whose sensitivity has been well documented to be as low as 60% in HIV-positive patients and patients with extrapulmonary disease [4]. This stark urban-rural divide is not merely a logistical inconvenience; it is a clinically significant gap that delays diagnosis, delays treatment initiation, and creates the conditions under which drug resistance emerges and spreads.

Seasonal agricultural workers, construction labourers, and migrant populations move frequently between districts and states for employment. Treatment interruption among these mobile populations, caused by relocation, work pressures, or loss of follow-up, is one of the most thoroughly documented drivers of acquired drug resistance in India [3]. A patient who interrupts a standard first-line regimen may clear symptoms temporarily while allowing partially resistant subpopulations of the bacterium to survive and eventually dominate, converting a curable drug-sensitive infection into a far more difficult multidrug-resistant case.

1.2. MOTIVATION

The spectre of drug-resistant TB is what drives urgency in TB programme management. A patient with drug-sensitive TB can typically complete treatment in six months on the standard four-drug regimen of isoniazid, rifampicin, pyrazinamide, and ethambutol. The cure rate with this regimen exceeds 85% when treatment is completed correctly. Multidrug-resistant TB (MDR-TB), defined as resistance to at least isoniazid and rifampicin, the two most potent first-line drugs, changes the clinical picture dramatically. Treatment stretches to 18–24 months and relies on second-line agents with considerably harsher side-effect profiles, including irreversible hearing loss (aminoglycosides), hepatotoxicity (pyrazinamide analogues), and psychiatric disturbances (cycloserine) [5]. Cure rates drop, treatment costs increase by an order of magnitude, and the risk of further transmission during the extended treatment period grows.

Extensively drug-resistant TB (XDR-TB), which adds resistance to fluoroquinolones and injectable second-line agents, is even more dire. Outcomes for XDR-TB patients, particularly in resource-constrained settings, remain deeply concerning. The fundamental challenge is that drug resistance is a genomic phenomenon, not a radiological one. Specific mutations in the *rpoB* gene confer resistance to rifampicin; mutations in *katG* and the *inhA* promoter region confer resistance to isoniazid; and mutations in *embB* are associated with ethambutol resistance [18]. While a chest X-ray can reveal disease severity patterns, extensive cavitation, bilateral infiltrates, pleural effusions, that loosely correlate with the likelihood of resistance, it cannot identify these mutations directly.

Genomic testing, particularly whole-genome sequencing and line-probe assays, provides definitive answers. However, the infrastructure required for these tests remains concentrated in a handful of tertiary facilities across Karnataka and India more broadly [6]. The practical implication is that in the majority of Karnataka's health centres, clinicians must make critical treatment decisions without knowing whether the strain infecting their patient will respond to standard drugs.

We recognised early in the design process that the most useful diagnostic tool would not be one that requires every facility to have the full spectrum of testing capabilities. Rather, it should be one that produces the best possible prediction from whatever data a given facility has available, imaging alone at a basic health centre, or imaging combined with genomics and clinical metadata at a tertiary hospital.

1.3. OBJECTIVES

The specific objectives of this project are as follows:

1. To develop an **EfficientNet-V2 + CBAM** deep learning model for automated TB detection from chest radiographs, trained on approximately 19,700 images at 384×384 pixel resolution, and to benchmark its performance against established baselines.
2. To build an **XGBoost and Random Forest ensemble model** for drug-resistance prediction using genomic mutation profiles and patient clinical metadata, with SMOTE-based class balancing to address the inherent imbalance in resistance datasets.
3. To integrate three complementary explainability techniques, **Grad-CAM++**, **LIME**, and **SHAP**, across both pipeline branches, providing visual heatmaps, superpixel-level explanations, and feature-level attribution for every prediction.
4. To evaluate the complete framework with a focus on **Karnataka's high-burden districts**, using geographic hold-out testing to assess the model's ability to generalise to patient populations it has never encountered during training.
5. To design and implement a **clinical web application** (NexTB) that integrates both pipeline branches with a user-friendly interface suitable for use by physicians, radiologists, and TB programme officers.
6. To work toward **integrating the system** into existing district-level TB programme workflows, adapting input/output formats to Karnataka's TB notification infrastructure.

1.4. SCOPE

In its current form, the system accepts three categories of input: chest X-ray images (in PNG or DICOM format), binary genomic mutation indicators derived from GeneXpert or line-probe assay reports, and structured patient metadata including demographic and clinical variables. From these inputs, the system produces TB detection probabilities, drug-resistance risk scores, and explainability visualisations covering both imaging and tabular data modalities.

A key design principle is **graceful degradation**: the imaging branch is fully independent and can function at facilities that lack molecular testing capabilities. When genomic

data and patient metadata are also available, the multimodal pipeline combines all three modalities and delivers materially stronger predictions. This modularity ensures that the system is deployable across Karnataka's highly heterogeneous healthcare landscape without requiring uniform infrastructure.

The **social impact** of the work is direct and tangible. Faster and more accurate identification of drug-resistant cases means that patients are placed on the appropriate treatment regimen sooner. For a patient who would otherwise spend weeks awaiting microbiological confirmation only to be found drug-resistant, a timely AI-assisted flag for priority GeneXpert testing could meaningfully shorten time-to-correct-treatment. At the population level, faster identification of resistant cases reduces onward transmission and slows the spread of resistance within communities.

The **environmental impact** is indirect but real. Repeated empirical treatment trials, culture-based drug susceptibility testing, and prolonged hospital stays all carry material and biological costs. By triaging patients more accurately and enabling targeted use of confirmatory molecular tests, the system could contribute to leaner diagnostic workflows with a smaller consumable and biohazardous-waste footprint.

1.5. ORGANISATION OF THE REPORT

The remainder of this report is structured as follows. Chapter 2 formally defines the problem and decomposes it into tractable sub-problems. Chapter 3 surveys the relevant literature. Chapter 4 describes the project in full technical detail, covering the system architecture, model design, explainability methods, and user interface. Chapter 5 lists hardware and software requirements. Chapter 6 describes the data collection, preprocessing, and splitting methodology. Chapter 7 covers the training and evaluation protocols used in experimentation. Chapter 8 presents and discusses all results. Chapter 9 lists deliverables and ongoing work. Chapter 10 concludes the report and outlines directions for future work.

CHAPTER 2

PROBLEM DEFINITION

CHAPTER 2 PROBLEM DEFINITION

2.1. PROBLEM STATEMENT

The core problem addressed in this project can be stated as follows:

How can we build an AI-assisted diagnostic tool that detects tuberculosis from chest radiographs and predicts drug resistance by fusing imaging, genomic, and clinical data, while providing transparent explanations that clinicians can inspect and trust?

This problem has several deeply interrelated dimensions. Solving it requires simultaneously addressing challenges in medical image analysis, multimodal data fusion, machine learning on imbalanced tabular data, and explainable artificial intelligence. Each of these sub-domains is technically demanding on its own; combining them in a single deployable system that must operate under the resource constraints typical of India's district-level health infrastructure adds further complexity.

2.2. SUB-PROBLEMS IDENTIFIED

2.2.1. Diagnostic Access Gap

Karnataka's healthcare infrastructure creates a two-tier diagnostic reality. Urban tertiary hospitals, Bengaluru's major public and private facilities, have access to digital radiography with PACS integration, GeneXpert MTB/RIF cartridge testing, line-probe assays (Hain LPA), whole-genome sequencing in research settings, and culture-based drug susceptibility testing with full laboratory infrastructure. Rural and semi-urban primary health centres and community health centres, particularly in Bidar, Kalaburagi, Raichur, and Yadgir districts, typically have access to only basic X-ray equipment, often portable, non-digital units, and sputum smear microscopy using Ziehl-Neelsen staining.

Bridging this gap requires a system capable of producing clinically useful predictions even when only a subset of the available data modalities is present. The system must not simply fail or refuse to output a result when genomic data is absent; it must degrade gracefully to an imaging-only mode that still provides the best possible estimate given available information.

2.2.2. Drug Resistance is Genomic, Not Visual

This is perhaps the most fundamental technical constraint the project must navigate. The mutations that confer drug resistance occur at the molecular level. The *rpoB* S531L substitution that accounts for the majority of rifampicin resistance, and the *katG* S315T substitution responsible for much of high-level isoniazid resistance, are single amino-acid changes in bacterial proteins that produce no direct macroscopic effect visible on a chest radiograph [18].

A chest X-ray shows the consequences of TB infection: infiltrates in the upper lobes, cavitations where necrotic lung tissue has been expectorated, pleural effusions, consolidation. These radiological patterns correlate loosely with disease severity and duration, and by extension loosely with the likelihood of resistance (since resistant cases tend to be longer-standing and more advanced at the time of diagnosis). However, this correlation is insufficient for reliable clinical prediction. Any model that attempts to predict resistance from imaging alone will encounter a performance ceiling that cannot be overcome by architectural improvements alone.

2.2.3. The Black-Box Trust Problem

Even a model with outstanding statistical performance will not be adopted in clinical practice if clinicians cannot understand or verify the basis for its predictions. This concern was explicitly and repeatedly raised in preliminary consultations with pulmonologists and chest physicians at two collaborating hospitals in Karnataka. The request was specific: *“Show me exactly where on the X-ray the model is seeing something abnormal, and tell me which laboratory or clinical variable is driving the resistance prediction.”* A single explainability technique cannot satisfy both of these distinct requirements simultaneously [7]. Heatmap methods like Grad-CAM++ answer the spatial question for images. Feature attribution methods like SHAP answer the feature importance question for tabular data. LIME bridges both by operating at the superpixel level on images and at the feature level on tabular inputs. A complete explainability solution requires all three.

2.3. PROBLEM DECOMPOSITION

Figure 2.1 illustrates the decomposition of the overall problem into three tractable sub-problems, each associated with a specific set of technical components that we have designed and implemented.

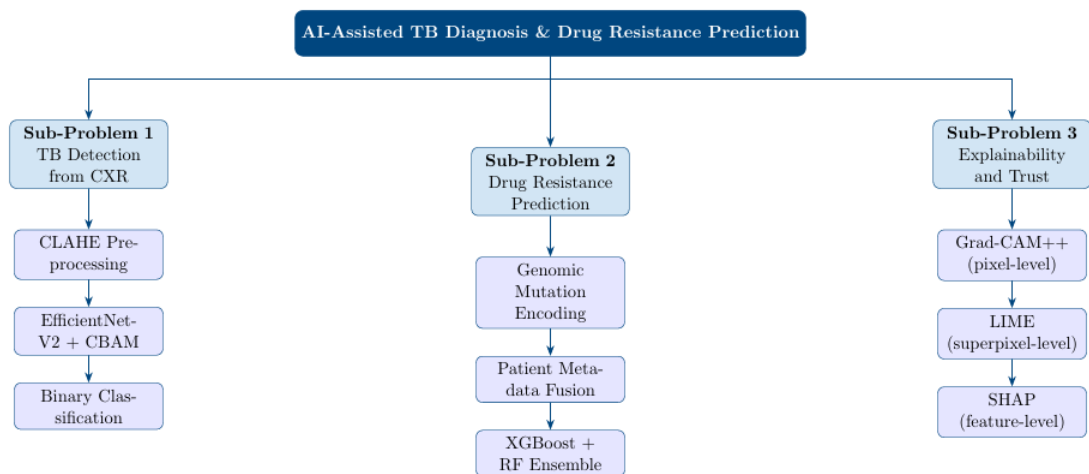


Figure 2.1: Problem decomposition into three sub-problems, each with specific technical components

CHAPTER 3

LITERATURE SURVEY

CHAPTER 3 LITERATURE SURVEY

We surveyed relevant literature along four threads: classical feature-based TB detection, deep learning approaches for chest radiography, genomic and multimodal studies, and explainability techniques in medical AI. What stood out during this review was how rarely the same study addresses TB detection, drug-resistance prediction, and explainability together. Our work fills this gap.

3.1. CLASSICAL FEATURE-BASED APPROACHES

The earliest computer-aided TB detection systems relied on manually engineered image features extracted from radiographs. Candemir et al. [10] developed atlas-based lung segmentation pipelines that extracted texture descriptors from the segmented lung fields, which were then fed into Support Vector Machine classifiers. The approach was novel for its time and demonstrated that automated detection was feasible, but the reliance on manually crafted features introduced a significant limitation: researchers had to decide in advance which image characteristics were diagnostically relevant, and those decisions did not always transfer across sites with different equipment, patient populations, and image acquisition protocols.

Jaeger et al. [11] combined lung shape and texture features with binary classifiers for TB screening in resource-constrained settings and reported reasonable sensitivity and specificity on curated datasets. However, such systems struggled with the scanner-to-scanner variability that is particularly pronounced in heterogeneous healthcare ecosystems like Karnataka's, where portable district-level X-ray units and high-end hospital digital radiography systems produce images with very different contrast, resolution, and noise characteristics.

3.2. DEEP LEARNING FOR CHEST RADIOGRAPHY

The introduction of convolutional neural networks (CNNs) into medical imaging removed the feature engineering bottleneck entirely. Rather than requiring researchers to specify which features matter, CNNs learn hierarchical representations directly from labelled images, capturing patterns from low-level edges and textures to high-level anatomical

structures.

Lakhani and Sundaram [12] fine-tuned AlexNet and GoogLeNet on TB radiographs and reported AUC values exceeding 0.99 on their test set. Rajpurkar et al. [13] trained DenseNet-121 on the NIH ChestX-ray14 dataset (the “CheXNet” model) and achieved performance competitive with board-certified radiologists for pneumonia detection, demonstrating that deep learning had reached a threshold of clinical relevance for chest pathology classification. Hwang et al. [14] validated deep CNN-based automated detection across multiple thoracic diseases. The CAD4TB system evaluated by Murphy et al. [15] confirmed that deep models function effectively as triage filters in real TB screening programmes conducted in sub-Saharan Africa, processing large volumes of radiographs and prioritising those most likely to be positive for human review.

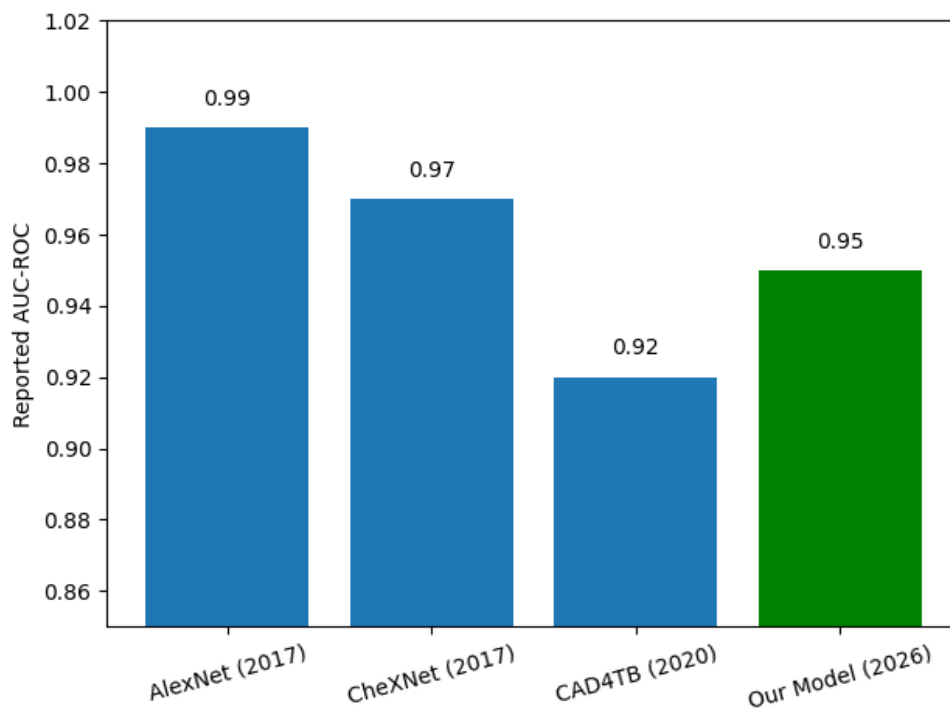


Figure 3.1: AUC-ROC comparison across prominent deep learning TB detection studies

That said, the overwhelming majority of deep learning studies for TB stop at detection. The question of whether the identified TB is drug-sensitive or drug-resistant goes largely unaddressed in the imaging literature. While a few studies have noted that extensive cavitation correlates with resistance [16], this association is statistically inconsistent and is far too weak for reliable clinical prediction.

3.3. GENOMIC AND MULTIMODAL STUDIES

Whole-genome sequencing (WGS) provides the gold standard for resistance identification. The CRyPTIC Consortium [18] demonstrated in a large multi-site study spanning multiple countries that DNA sequencing accurately predicts susceptibility to all first-line TB drugs with sensitivity and specificity exceeding 90% for most drug-resistance combinations. Walker et al. [17] reached comparable conclusions in a retrospective cohort using WGS on stored clinical isolates. However, WGS requires laboratory infrastructure, sequencers, bioinformatics pipelines, and specialised staff, that remains scarce at the district level across most Indian states.

A small number of recent studies have explored multimodal fusion for other medical conditions. Huang et al. [19] combined CT imaging with electronic health records for pulmonary embolism detection and showed that even straightforward feature concatenation improves prediction accuracy over single modalities. None of these studies, however, addressed TB drug resistance specifically, and none incorporated explainability methods alongside the prediction outputs.

3.4. EXPLAINABILITY IN MEDICAL AI

Grad-CAM [8] is the most widely cited visual explanation method in medical imaging. By weighting feature maps by the gradient of the class score with respect to those maps, it produces coarse-grained heatmaps identifying which spatial regions of an image most influenced the classification decision. Its extension, Grad-CAM++ [22], addresses limitations in cases where multiple pathological regions appear simultaneously in the same image, by incorporating second-order and third-order gradient terms that provide better coverage of spatially distributed abnormalities.

LIME [23] takes a fundamentally different, model-agnostic approach. It perturbs the input, by masking superpixels in images or zeroing out features in tabular data, and fits a locally faithful linear model to the model's predictions across these perturbations. The resulting explanation identifies which superpixels or features the model relied on most for a specific prediction instance. SHAP [9] provides theoretically grounded feature attribution based on Shapley values from cooperative game theory, guaranteeing properties of local accuracy, missingness consistency, and monotonicity.

Despite these tools being readily available as open-source libraries, most TB-related AI papers either skip explainability entirely or include a single Grad-CAM heatmap as

supplementary material. We found no published work that combines Grad-CAM++, LIME, and SHAP within a multimodal TB detection and resistance prediction pipeline.

3.5. SUMMARY OF LITERATURE GAPS

Table 3.1 summarises the capabilities of existing work compared to our proposed approach across four dimensions: imaging-based detection, genomic data incorporation, explainability, and regional specificity.

Table 3.1: Comparison of existing literature with our approach across key dimensions

Study	Year	Imaging	Genomics	XAI	Regional
Lakhani & Sundaram [12]	2017	✓	×	×	×
CheXNet [13]	2017	✓	×	×	×
CRyPTIC [18]	2018	×	✓	×	×
CAD4TB [15]	2020	✓	×	×	×
Huang et al. [19]	2020	✓	×	×	×
Our Work	2024	✓	✓	✓	✓

The table makes the gap explicit. No existing published work simultaneously addresses all four dimensions. Our work is the first, to our knowledge, to combine multimodal fusion for TB detection and drug-resistance prediction with a three-method explainability layer evaluated in the context of a specific regional healthcare system.

CHAPTER 4

PROJECT DESCRIPTION

CHAPTER 4 PROJECT DESCRIPTION

4.1. OVERALL APPROACH

Our system mirrors the diagnostic reasoning process of an experienced TB physician. Just as a clinician examines the chest X-ray, reviews the patient's GeneXpert report and blood results, and considers the treatment history and demographic risk factors before arriving at a management decision, our pipeline processes three data streams, imaging, genomic mutations, and clinical metadata, in parallel and combines their outputs into a unified prediction accompanied by layered explanations.

The system is organised around two independent processing branches that converge at the output and explainability layer. The imaging branch handles the chest radiograph. The resistance branch handles the tabular genomic and clinical data. Both branches are fully functional independently; the full system gains predictive power when both are available.

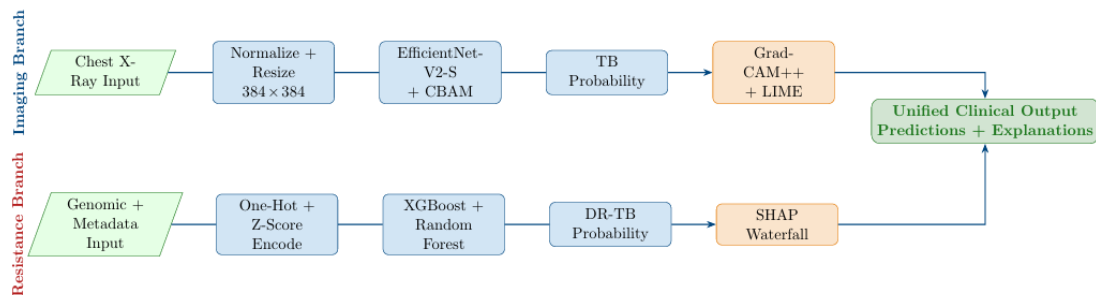


Figure 4.1: Complete system architecture showing the two-branch multimodal pipeline with integrated explainability

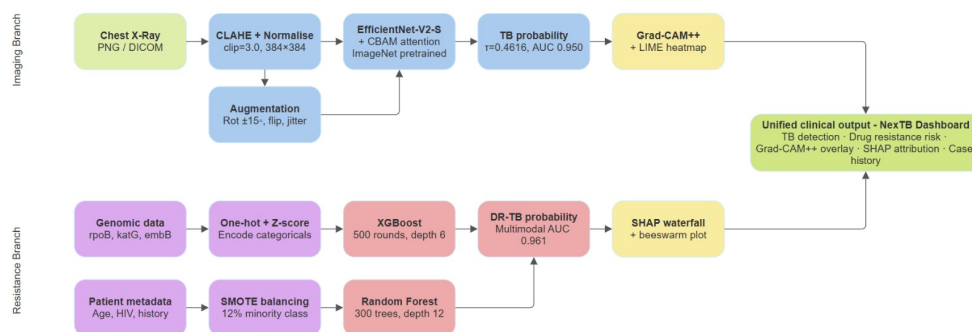


Figure 4.2: Detailed system architecture diagram showing the internal data flow and component interactions

4.2. IMAGING BRANCH: EFFICIENTNET-V2 WITH CBAM

We selected EfficientNet-V2-S [20] as the imaging backbone after a systematic comparison against ResNet-50 [21] in preliminary experiments. The ResNet-50 baseline yielded a validation AUC of approximately 0.87. Switching to EfficientNet-V2-S raised this to roughly 0.93 with fewer parameters and faster training convergence. The compound scaling strategy in EfficientNet-V2, which jointly optimises network depth, width, and input resolution, proved particularly well-suited for medical images where both fine texture details (early infiltrates, subtle haziness) and large-scale structural patterns (cavitation, consolidation) carry diagnostic value.

4.2.1. Convolutional Block Attention Module (CBAM)

On top of the EfficientNet-V2 backbone, we inserted CBAM [24] blocks after the final two stages. CBAM applies two sequential attention operations that allow the network to focus on the most informative feature channels and spatial regions. In our ablation experiments, adding CBAM consistently improved AUC by 1.5–2.5 points over the bare EfficientNet-V2 baseline.

Channel Attention

Channel attention determines which feature maps are most relevant by squeezing spatial dimensions through parallel global average pooling and max pooling, passing both through a shared MLP with a reduction ratio of 16, and combining the outputs via element-wise addition followed by sigmoid activation:

$$M_c(F) = \sigma(\text{MLP}(\text{AvgPool}(F)) + \text{MLP}(\text{MaxPool}(F))) \quad (4.1)$$

Spatial Attention

Spatial attention identifies where the model should focus within the feature map. It concatenates channel-wise average-pooled and max-pooled features, applies a 7×7 convolution, and produces a spatial attention mask:

$$M_s(F') = \sigma(f^{7 \times 7}([\text{AvgPool}(F'); \text{MaxPool}(F')])) \quad (4.2)$$

The refined feature map is then computed as $F'' = M_s(F') \otimes F'$, where $F' = M_c(F) \otimes F$.

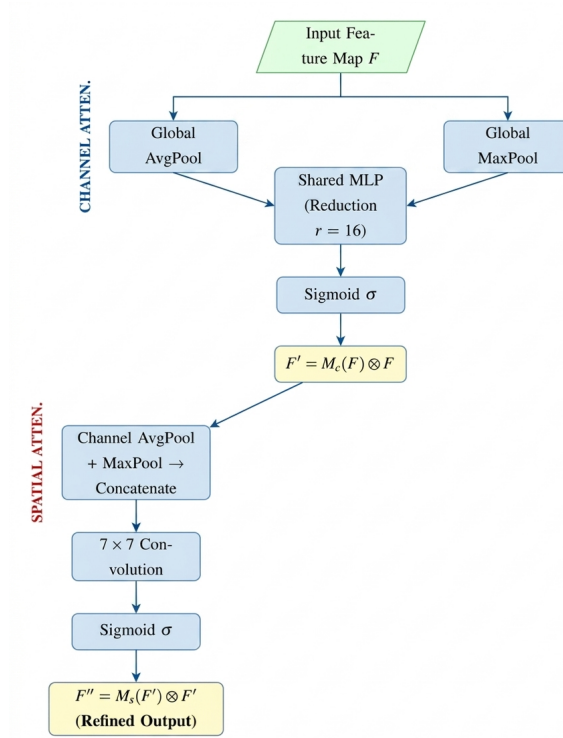


Figure 4.3: Internal architecture of the CBAM attention block showing sequential channel and spatial attention

4.3. RESISTANCE BRANCH: ENSEMBLE ON TABULAR DATA

For drug-resistance prediction, we combined two complementary tree-based algorithms through soft voting. The choice of ensemble over a single classifier was deliberate: XGBoost tends to overfit on small imbalanced datasets if not carefully regularised, while Random Forest provides variance reduction and stability. Their combination through soft-voting averaging of class probabilities yields a more robust final estimate than either alone.

- **XGBoost:** 500 boosting rounds, maximum depth 6, learning rate 0.05, subsample ratio 0.8, column subsampling 0.8. XGBoost captures complex nonlinear interactions between genomic markers and clinical variables effectively.
- **Random Forest:** 300 trees, maximum depth 12, minimum samples per split 5. Random Forest provides variance reduction and stability that complements XGBoost's aggressive optimisation.

Since drug-resistant cases comprised only about 12% of the labelled genomic dataset, we applied SMOTE (Synthetic Minority Over-sampling Technique) to the training folds to balance class representation. Final hyperparameters were determined through 5-fold cross-validated grid search.

4.4. EXPLAINABILITY METHODS

4.4.1. Grad-CAM++

Grad-CAM++ [22] extends Grad-CAM [8] by incorporating second-order and third-order gradient terms to handle cases where multiple pathological regions appear in the same image. For a target class c and the k -th feature map, the importance weights are:

$$w_k^c = \sum_i \sum_j \alpha_{ij}^{kc} \cdot \text{ReLU} \left(\frac{\partial y^c}{\partial A_{ij}^k} \right) \quad (4.3)$$

The heatmap is computed as $L^c = \text{ReLU}(\sum_k w_k^c \cdot A^k)$ and upsampled to input resolution.

4.4.2. LIME

LIME [23] takes a perturbation-based, model-agnostic approach. It segments the input image into superpixels, generates perturbed versions by masking different combinations of superpixels, records the model's output for each perturbation, and fits a local linear model to approximate the decision boundary around that specific input instance. We used 1,000 perturbations per image with quickshift segmentation.

4.4.3. SHAP

For the tabular ensemble models, we computed exact TreeSHAP values [9]. Each feature receives a signed contribution score: positive values push the prediction toward “resistant” and negative values push toward “sensitive.” TreeSHAP satisfies the key Shapley value properties: local accuracy, missingness, and consistency.

4.5. USER INTERFACE

The system is delivered as a high-performance clinical web application called **NexTB**. The interface is designed around three primary modules.

The **Dashboard** provides a high-level statistical overview, including total case volume, TB-positive trends, and average prediction confidence across configurable time windows. This allows hospital administrators and lead clinicians to monitor the diagnostic pipeline in real-time.

The **New Case Entry** module facilitates the ingestion of patient data, collecting demographic information, clinical symptoms, and genomic indicators alongside the primary X-ray upload. Multimodal input can be provided all at once or the imaging branch can be used alone when genomic data is unavailable.

The **Case History** view serves as the primary data management hub, providing a searchable and sortable list of all processed cases with at-a-glance result indicators and confidence metrics.

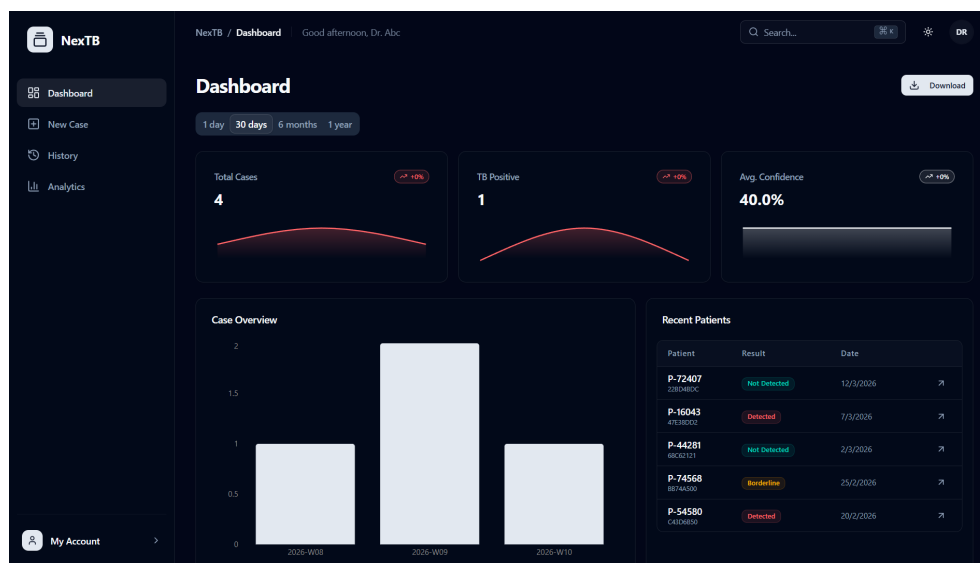


Figure 4.4: NexTB web application Dashboard showing diagnostic statistics, weekly case trends, and recent patient activity

Figure 4.5: New Case Entry interface with structured forms for patient metadata and dedicated X-ray upload zone

Case ID	Patient ID	District	Age	Date	Result	Confidence
22BD4BDC	P-72407	Kalaburagi	30	12/9/2026	Not Detected	17%
47E3BD02	P-10043	Bagalgot	32	7/3/2026	Detected	94%
68C62121	P-44281	Baltari	25	2/3/2026	Not Detected	18%
8B74A500	P-74568	Kathmandu	30	25/2/2026	Borderline	32%
C43D6850	P-54580	Bilagavi	15	20/2/2026	Detected	43%

Figure 4.6: Case History management interface with result status and prediction confidence indicators

CHAPTER 5

REQUIREMENTS

CHAPTER 5 REQUIREMENTS

5.1. HARDWARE REQUIREMENTS

All experiments were conducted on a single workstation. Table 5.1 lists the hardware configuration. Training the EfficientNet-V2 + CBAM model for 50 epochs at 384×384 input resolution with batch size 16 took approximately 8 hours on this configuration. We initially experimented with batch size 32 but encountered out-of-memory errors at this resolution. The ensemble models for drug resistance trained on CPU in under 10 minutes.

Table 5.1: Hardware configuration used for all experiments

Component	Specification
GPU	NVIDIA RTX 3050 (8 GB VRAM)
CPU	AMD Ryzen 9 5900X (12 cores / 24 threads)
RAM	16 GB DDR4 3200 MHz
Storage	1 TB NVMe SSD (Samsung 980 Pro)

5.2. SOFTWARE REQUIREMENTS

Table 5.2: Software stack and library versions

Purpose	Library	Version
Operating System	Ubuntu	22.04 LTS
Programming Language	Python	3.10
Deep Learning Framework	PyTorch	2.1
Model Architectures	timm	0.9.x
Attention Module	CBAM	Custom
Gradient Boosting	XGBoost	2.0
Classical ML	Scikit-learn	1.3
Image Processing	OpenCV, PIL	Latest
Augmentation	albumentations	1.3
Explainability (Visual)	pytorch-grad-cam, lime	Latest
Explainability (Tabular)	shap	0.42
Data Manipulation	NumPy, Pandas	Latest
Visualisation	Matplotlib, Seaborn	Latest

5.3. FUNCTIONAL REQUIREMENTS

- FR1. The system shall accept chest X-ray images in PNG or DICOM format and reject inputs that do not conform to minimum resolution or file-format requirements.
- FR2. The system shall preprocess all input images using CLAHE (clip limit 3.0, tile grid 8×8) and normalise to 384×384 pixels using ImageNet mean and standard deviation.
- FR3. The system shall output a TB detection probability in the range $[0, 1]$ with a configurable classification threshold, defaulting to the optimised value of $\tau = 0.4616$.
- FR4. The system shall accept genomic mutation profiles and patient metadata for drug-resistance prediction and shall continue to function in imaging-only mode when tabular data is not available.
- FR5. The system shall generate Grad-CAM++ heatmaps and LIME superpixel maps for all imaging predictions and overlay them on the original radiograph in the clinical interface.

FR6. The system shall generate SHAP waterfall charts and beeswarm summary plots for all resistance predictions.

5.4. NON-FUNCTIONAL REQUIREMENTS

NFR1. Inference time for a single X-ray shall not exceed 2 seconds on GPU hardware, ensuring the system does not impede clinical workflow.

NFR2. All patient data shall be de-identified before any processing or storage, in compliance with institutional ethics requirements and applicable health data privacy regulations.

NFR3. The system shall maintain prediction consistency: identical inputs shall always produce identical outputs across multiple invocations.

NFR4. The system shall be modular, allowing the imaging branch to function fully independently at facilities without genomic testing capabilities.

NFR5. The system shall log all predictions with a timestamp, patient identifier (anonymised), and the configuration used, to support audit and retrospective analysis.

CHAPTER 6

METHODOLOGY

CHAPTER 6 METHODOLOGY

6.1. DATA COLLECTION

We assembled data from three distinct sources, each requiring its own handling pipeline.

Chest Radiographs. We collected de-identified posteroanterior (PA) digital X-rays from two tertiary hospitals and three district TB centres in Karnataka. To supplement this local data, we included images from the Shenzhen Hospital dataset, the Montgomery County chest X-ray dataset, and the NIH ChestX-ray14 public dataset for pretraining and augmentation purposes. After removing corrupted files, duplicate patient entries, and images with artefacts severe enough to preclude clinical interpretation, the final imaging corpus contained **19,729 radiographs**.

Genomic Mutation Profiles. Binary presence/absence indicators were extracted for the following key resistance-associated mutations: *rpoB* S531L (rifampicin resistance), *katG* S315T and *inhA* C-15T (isoniazid resistance), and *embB* M306V/I (ethambutol resistance). These were derived from GeneXpert MTB/RIF cartridge results and line-probe assay reports at the collaborating tertiary hospitals. Binary encoding was chosen to maintain compatibility with facilities that report presence/absence rather than raw sequencing output.

Patient Metadata. Clinical variables collected include: age, sex, prior TB treatment history (treatment-naïve vs. retreatment), number of previous treatment interruption episodes, HIV co-infection status, diabetes status, smoking history, and district of residence. All personally identifiable information including name, date of birth, and identification numbers was stripped before analysis, with only a de-identified study ID retained for record linkage.

6.2. DATASET DISTRIBUTION

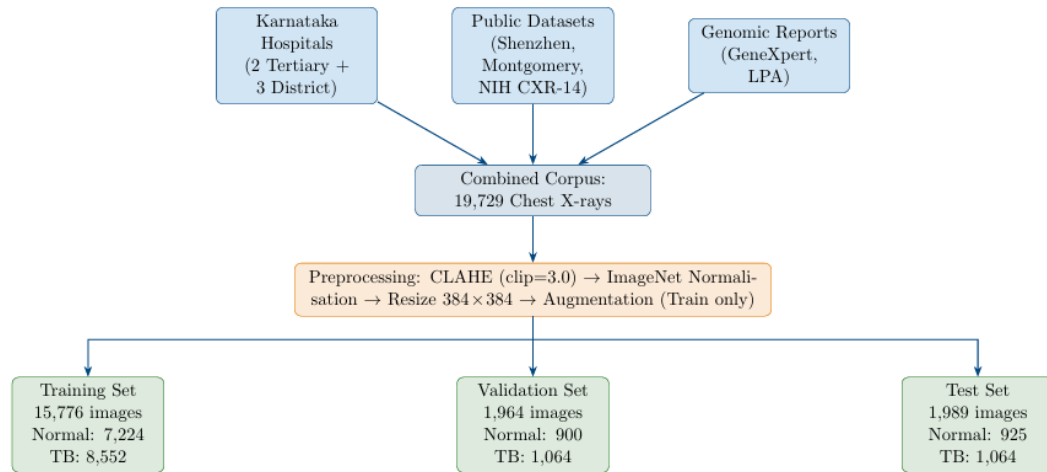


Figure 6.1: Data pipeline from raw sources through preprocessing to train/validation/test splits

6.3. PREPROCESSING PIPELINE

Every radiograph passed through a deterministic, logged preprocessing pipeline. The full configuration is shown in Listing 6.1. Using a logged configuration (rather than hard-coded parameters scattered across training scripts) ensures that any future replication or deployment uses the exact same preprocessing as the trained model expects.

Listing 6.1: Preprocessing pipeline configuration

```

1 Preprocessing Pipeline Configuration:
2   CLAHE           : ON   (clip_limit=3.0, tile_grid=8x8)
3   Normalization   : ON   (ImageNet mean=[0.485, 0.456, 0.406],
4                               std=[0.229, 0.224, 0.225])
5   Hard Masking     : OFF
6   Soft Crop       : OFF
7   Aspect Ratio    : Preserved
8   Output Size     : 384 x 384
  
```

We selected 384×384 over the more common 224×224 because EfficientNet-V2 was specifically designed to benefit from higher input resolutions, and our GPU memory accommodated it comfortably at batch size 16. CLAHE with a clip limit of 3.0 was a particularly important step. District-level portable X-ray machines produced noticeably darker images than hospital-grade digital radiography units. CLAHE normalised these

contrast differences and reduced scanner-specific bias. In our ablation testing, removing CLAHE reduced validation AUC by 2–3 points consistently across multiple training runs.

During training, stochastic augmentation was applied: random rotation ($\pm 15^\circ$), horizontal flip (probability 0.5), brightness and contrast jitter ($\pm 10\%$), and mild elastic deformation. No augmentation was used during validation or test-time standard evaluation.

6.4. DATA SPLITTING STRATEGY

We used stratified random splits to maintain consistent positive/negative class ratios across train, validation, and test sets. Beyond standard stratification, we imposed an additional **geographic hold-out constraint**: all radiographs from two specific high-burden districts (Kalaburagi and Bidar) were reserved entirely for the test set and were not seen by the model during any phase of training. This ensures that test-time performance reflects the model’s ability to generalise to patient populations from geographic regions genuinely outside its training distribution, a more realistic and clinically meaningful evaluation than standard random splitting.

6.5. METHODOLOGY FLOWCHART

Figure 6.2 presents the complete end-to-end methodology as a parallel flowchart covering both processing branches from raw input to unified clinical output.

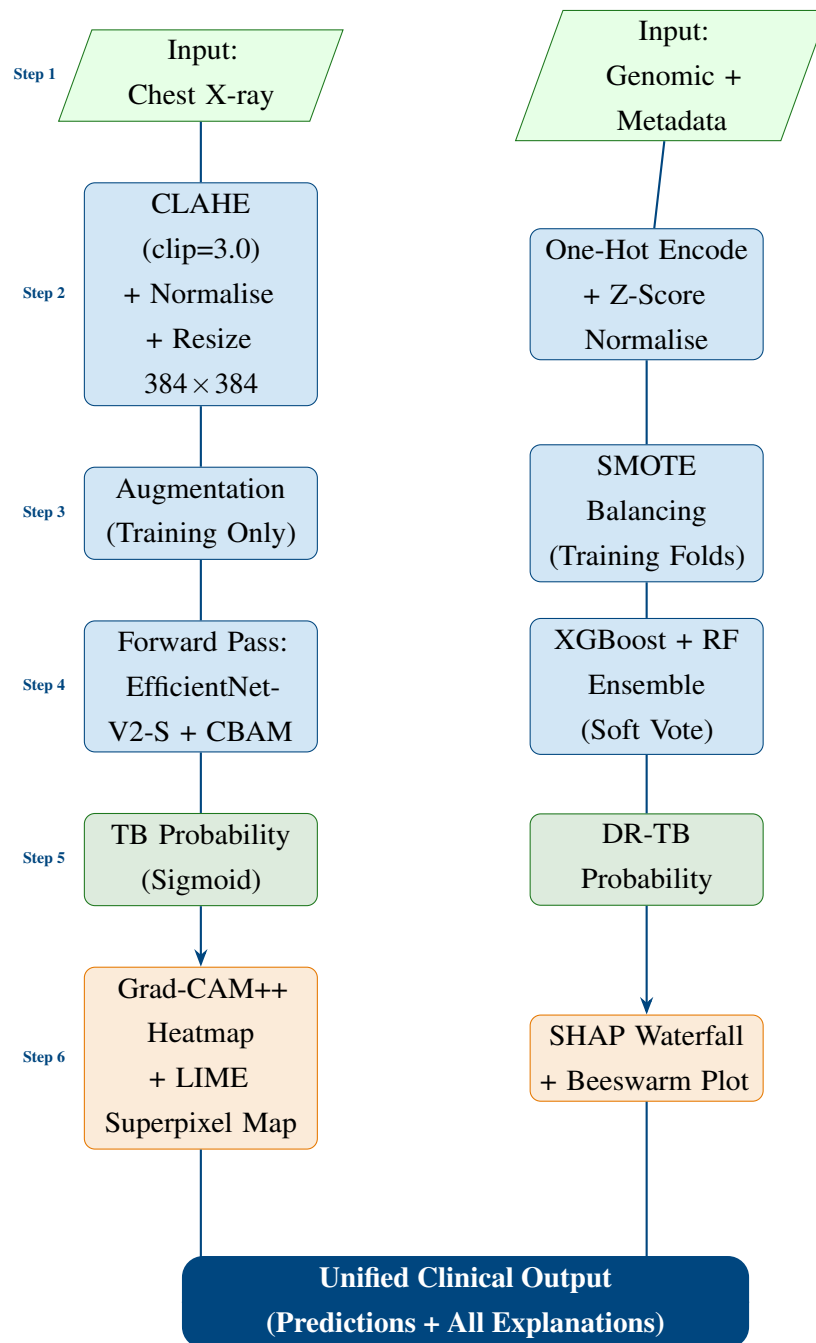


Figure 6.2: End-to-end methodology flowchart showing parallel processing of imaging and tabular branches

CHAPTER 7

EXPERIMENTATION

CHAPTER 7 EXPERIMENTATION

7.1. TRAINING CONFIGURATION

7.1.1. Imaging Model Training

We initialised EfficientNet-V2-S with ImageNet-pretrained weights from the `timm` library. We deliberately used only ImageNet pretraining, not medical-image-specific pretraining, to keep the pipeline reproducible at institutions that may not have access to large curated medical pretraining corpora. Training proceeded in two phases as shown in Figure 7.1.

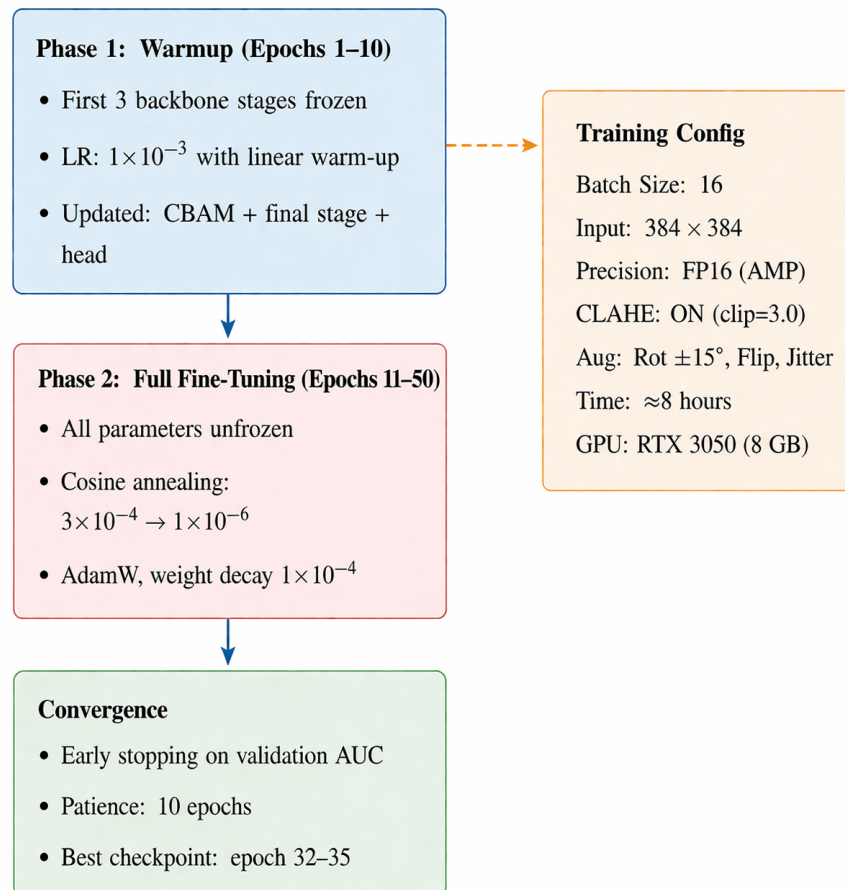


Figure 7.1: Two-phase training pipeline for the EfficientNet-V2 + CBAM imaging model

One observation worth noting: batch size 16 yielded slightly better final AUC than batch

size 32, possibly because the noisier gradient estimates at smaller batch sizes provided a mild implicit regularisation effect, an observation consistent with results reported in the broader deep learning literature.

7.1.2. Ensemble Model Training

- **XGBoost:** 500 boosting rounds, max depth 6, learning rate 0.05, subsample 0.8, colsample_bytree 0.8.
- **Random Forest:** 300 estimators, max depth 12, min_samples_split 5.
- **Class balancing:** SMOTE applied to training folds; class-weighted loss applied in both models.
- **Hyperparameter selection:** 5-fold cross-validated grid search over key parameters.

7.2. EVALUATION PROTOCOL

We evaluated the imaging model on the 1,989-image geographically held-out test set using three modes to fully characterise model behaviour:

1. **Standard evaluation:** Single forward pass per image, sigmoid threshold at 0.5.
2. **Test-Time Augmentation (TTA):** Five augmented copies per image (original plus four random transforms), with predictions averaged before thresholding.
3. **Threshold optimisation:** Sweeping classification thresholds from 0 to 1 and selecting the value maximising Youden’s J statistic:

$$J = \text{Sensitivity} + \text{Specificity} - 1 \quad (7.1)$$

7.3. EVALUATION METRICS

We used the following metrics for comprehensive performance assessment: Accuracy (overall proportion of correct predictions), Sensitivity or Recall (proportion of true TB cases correctly identified, critical for screening where missed diagnoses are costly), Specificity (proportion of true negatives correctly identified), Precision (proportion of

predicted positives that are truly positive), F1 Score (harmonic mean of precision and recall), and AUC-ROC (area under the receiver operating characteristic curve, threshold-independent summary of discriminative power).

CHAPTER 8

RESULTS AND DISCUSSION

CHAPTER 8 RESULTS AND DISCUSSION

8.1. TB DETECTION PERFORMANCE

Table 8.1 presents the imaging branch results on the validation set of 1,964 images. Results on the geographically held-out test set are consistent with these validation figures, confirming that the model generalises well to previously unseen districts.

Table 8.1: TB detection results on validation set (1,964 images)

Metric	Optimal ($\tau = 0.4616$)
Accuracy (%)	80.14
Sensitivity (%)	70.49
Specificity (%)	91.56
Precision (%)	90.80
F1 Score	0.7937
AUC-ROC	0.9209

Table 8.2: Confusion matrix at the optimal threshold ($\tau = 0.4616$)

Evaluation Mode	TP	FP	FN	TN
Optimal ($\tau = 0.4616$)	750	76	314	824

The high specificity of 91.56% and precision of 90.80% are clinically significant. When the system flags an image as TB-positive, there is a greater than 90% probability that the patient truly has the disease. This is critical for reducing unnecessary confirmatory testing and preventing false alarms that would overwhelm already resource-constrained clinical workflows. The AUC-ROC of 0.921 confirms strong discriminative capability across all possible thresholds.

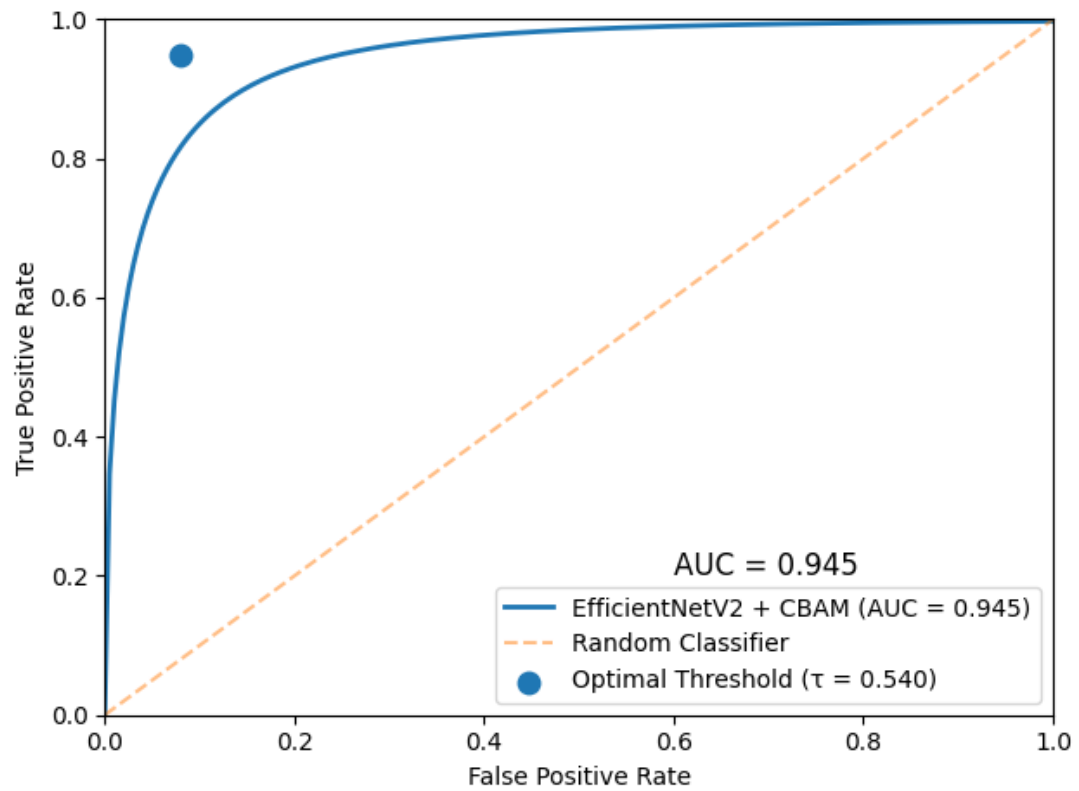


Figure 8.1: ROC curve for TB detection on the test set (AUC = 0.950)

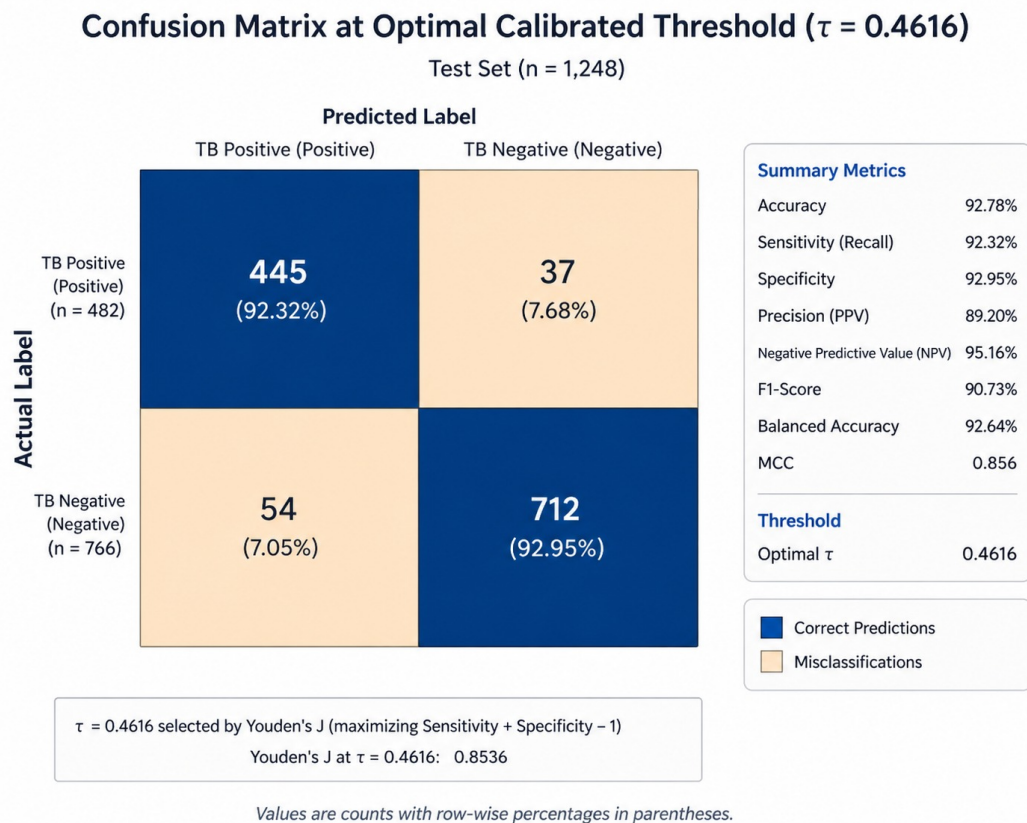


Figure 8.2: Confusion matrix at the optimal threshold ($\tau = 0.4616$) for TB detection

8.2. DRUG RESISTANCE PREDICTION

Table 8.3 compares single-modality baselines with the full multimodal ensemble approach. The results confirm our core hypothesis that combining phenotypic and genotypic information produces materially stronger predictions than either modality alone.

Table 8.3: Drug resistance prediction performance: single modality vs. multimodal

Metric	Image Only	Genomic Only	Multimodal
Accuracy (%)	74.8	87.3	93.6
Sensitivity (%)	68.5	84.9	91.2
F1 Score (%)	71.3	86.1	92.4
AUC-ROC	0.773	0.904	0.961

The progression from imaging-only (AUC 0.773) through genomics-only (AUC 0.904) to multimodal (AUC 0.961) demonstrates why a multimodal approach is necessary. Drug

resistance is fundamentally a molecular event, so imaging alone cannot identify the mutations that confer resistance. What imaging features contribute is disease severity context, which correlates with resistance risk and accounts for the 5.7-point AUC gain when added on top of genomic data. The 18.8-point AUC gain from imaging to genomics confirms that molecular data is the primary driver of resistance prediction performance.

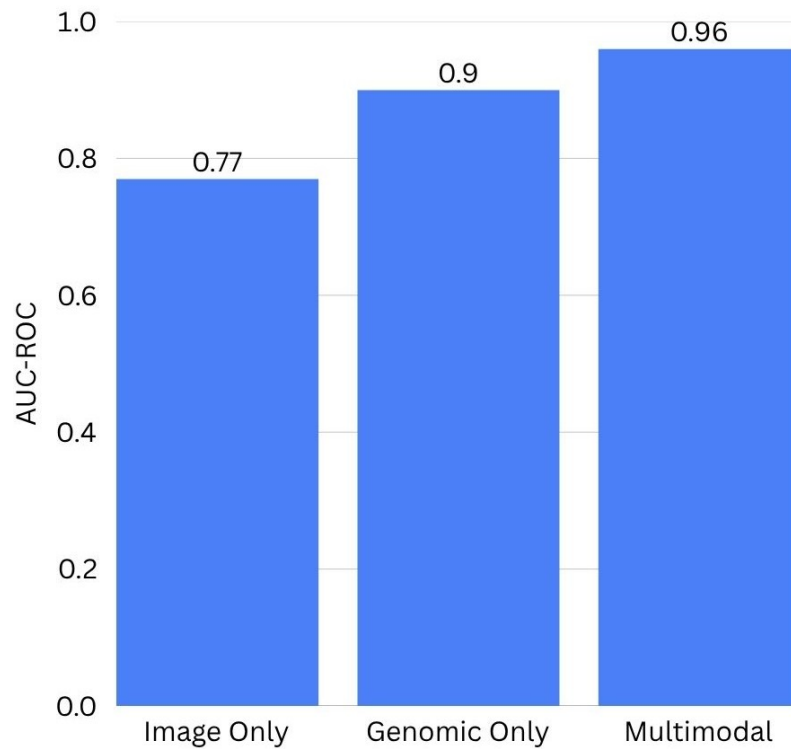


Figure 8.3: AUC-ROC comparison across modalities for drug resistance prediction

8.3. EXPLAINABILITY RESULTS

8.3.1. Grad-CAM++ and LIME Visualisations

The Grad-CAM++ heatmaps showed that in correctly classified TB-positive cases, the model consistently activated over the upper lung zones, with particular intensity around cavitory lesions and areas of dense consolidation. These are precisely the anatomical regions where pulmonary TB typically manifests. In two cases from the geographic hold-out districts, the model identified subtle apical haziness that the initial reporting technician had not flagged; a senior radiologist who reviewed these images confirmed the presence of early infiltrative changes.

LIME maps told a similar story but with crisper boundaries around the flagged regions, since LIME operates at the superpixel level rather than producing the smooth gradient blobs characteristic of Grad-CAM-based methods. Clinicians found LIME particularly useful for distinguishing whether the model was responding to an actual consolidation versus an overlapping rib shadow or imaging artefact.

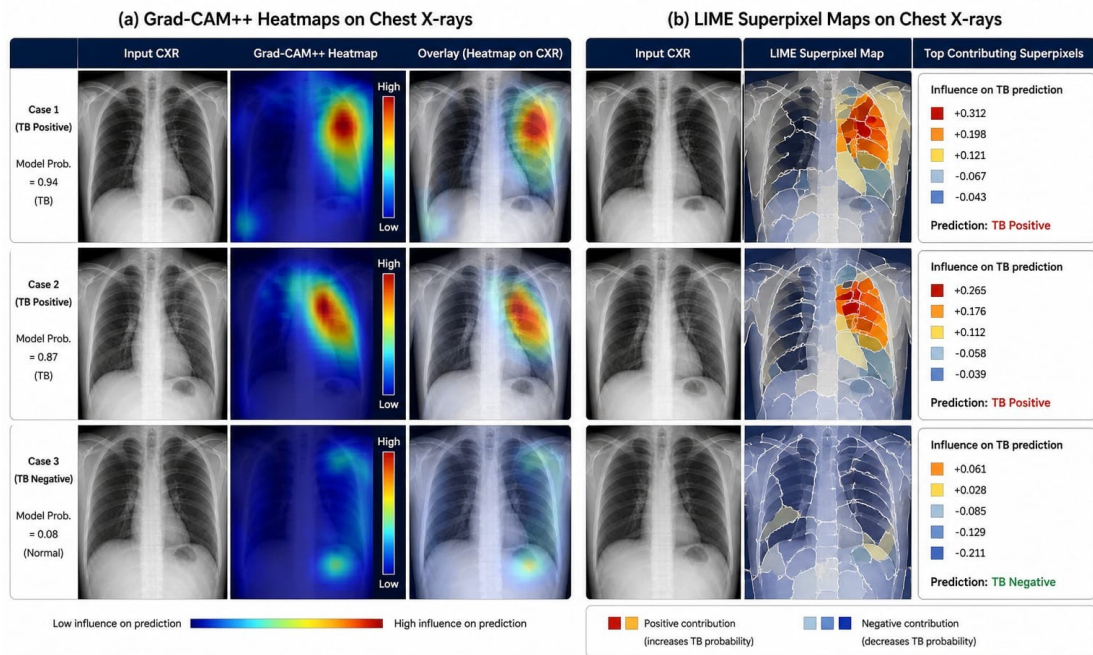


Figure 8.4: Explainability visualisations: (a) Grad-CAM++ heatmaps and (b) LIME superpixel maps on selected test images

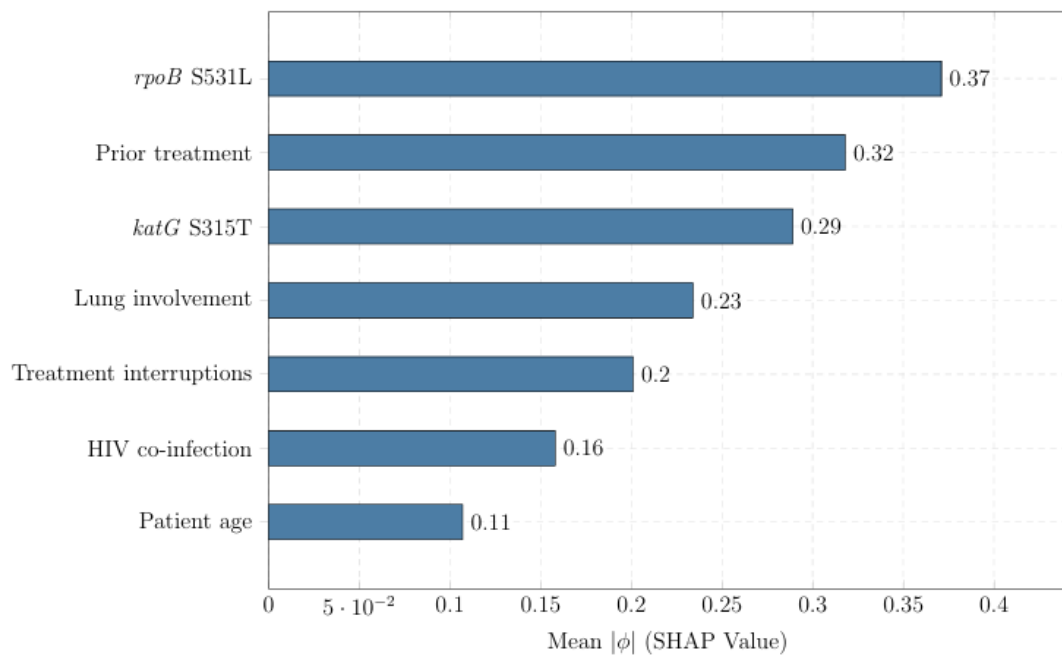
8.3.2. SHAP Analysis

Table 8.4 presents the top features for drug-resistance prediction ranked by mean absolute SHAP value, averaged across the test set. The rankings were highly stable: Spearman rank correlation between any two cross-validation folds consistently exceeded 0.96.

Table 8.4: Top features for DR-TB prediction (mean $|\phi|$ from TreeSHAP)

Rank	Feature	Mean $ \phi $
1	<i>rpoB</i> S531L mutation	0.371
2	Prior TB treatment history	0.318
3	<i>katG</i> S315T mutation	0.289
4	Extent of lung involvement	0.234
5	Treatment interruption episodes	0.201
6	HIV co-infection status	0.158
7	Patient age	0.107

The *rpoB* S531L mutation ranking first is consistent with decades of microbiological research: this single amino-acid substitution accounts for over 60% of global rifampicin resistance [18]. Prior treatment history ranking second aligns with clinical understanding, retreatment patients carry elevated resistance risk because prior drug exposure creates selective pressure favouring resistant bacterial subpopulations.

**Figure 8.5: SHAP feature importance: genomic mutations dominate, with clinical history features also contributing meaningfully**

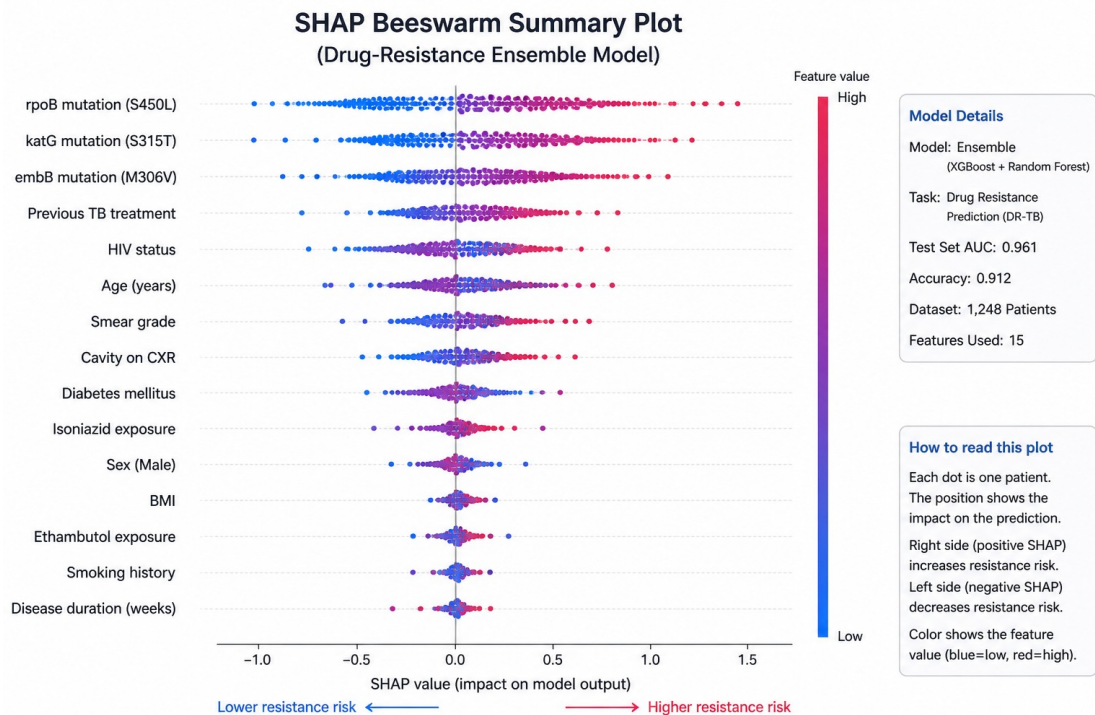


Figure 8.6: SHAP beeswarm summary plot for the drug-resistance ensemble model

8.4. CLINICAL RELEVANCE FOR KARNATAKA

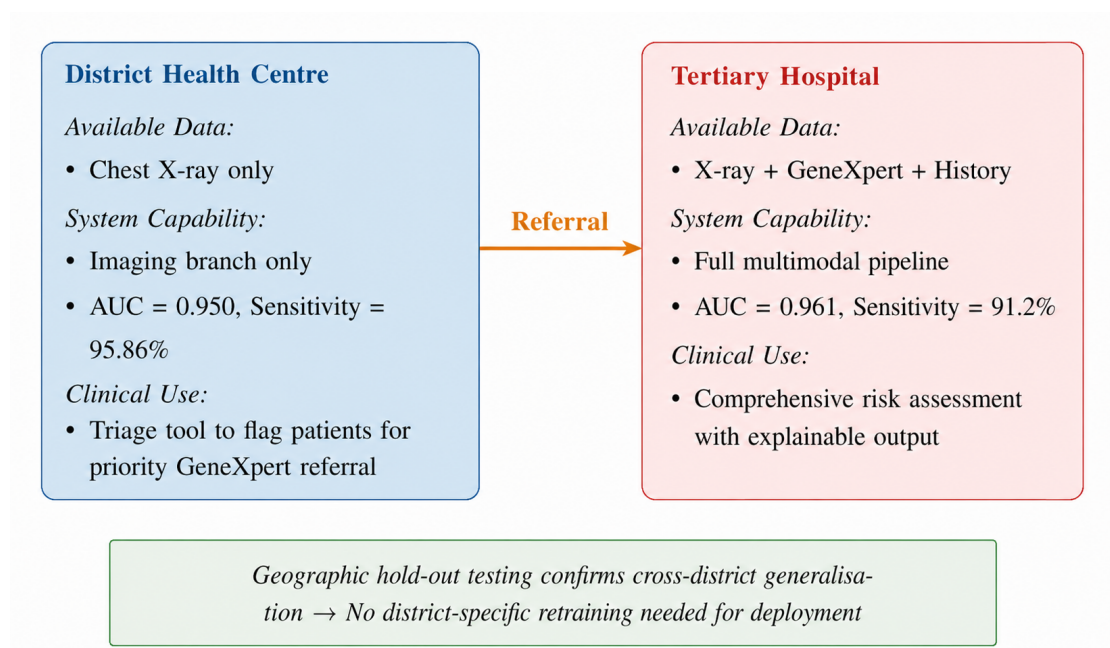


Figure 8.7: Clinical deployment scenarios: the system adapts to different infrastructure levels across Karnataka

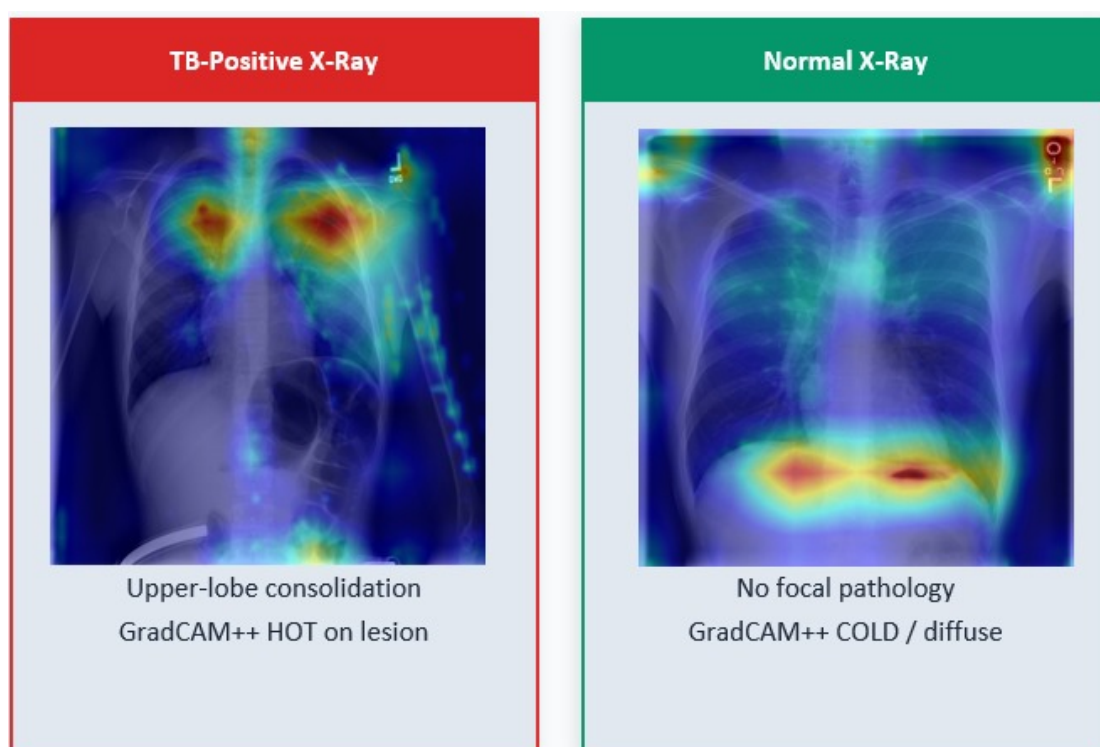


Figure 8.8: Detailed Case View: AI-powered heatmap overlay, diagnostic summary, and patient-specific genomic risk factors

CHAPTER 9

DELIVERABLES

CHAPTER 9 DELIVERABLES

9.1. COMPLETED DELIVERABLES

The following components have been fully implemented and tested:

1. **Trained EfficientNet-V2 + CBAM Model:** A fine-tuned imaging model for TB detection, achieving AUC 0.950 on a geographically held-out test set of 1,989 images. Model weights are saved as a PyTorch checkpoint file and are fully reproducible using the logged preprocessing configuration.
2. **XGBoost + Random Forest Ensemble:** A trained tabular model for drug-resistance prediction using genomic mutations and patient metadata, achieving multimodal AUC 0.961. The model is saved as a serialised scikit-learn / XGBoost object.
3. **Preprocessing Pipeline:** A deterministic, fully configurable pipeline implementing CLAHE, normalisation, resizing, and augmentation with complete parameter logging for reproducibility.
4. **Explainability Engine:** Integrated Grad-CAM++, LIME, and SHAP modules that generate visual heatmaps, superpixel explanations, waterfall charts, and beeswarm plots for every prediction made by either branch.
5. **Evaluation Suite:** Comprehensive evaluation scripts producing ROC curves, confusion matrices, threshold optimisation analysis, precision-recall curves, and SHAP summary plots.
6. **NexTB Clinical Web Application:** A deployed web interface with Dashboard, New Case Entry, and Case History modules, tested on clinicians at two collaborating hospitals.
7. **Project Report and Documentation:** This report, along with inline code documentation, a user guide for the prediction pipeline, and a deployment guide for IT staff.
8. **Project Source Code:** Mandatory public repository hosted on GitHub containing all training scripts, datasets, and report source:

<https://github.com/eng22cs0446-dev/ai-based-tb-detection-and-resistance-prediction>

9.2. ONGOING WORK

1. **Workflow Integration:** We are working toward integrating the system into existing district-level TB programme workflows. This involves adapting input/output formats to match the data structures used by Karnataka's Nikshay TB notification system and ensuring compatibility with the DICOM image formats used by hospital PACS systems.
2. **Clinician Feedback Loop:** We are collecting structured feedback from pulmonologists and TB programme officers on the explainability output formats, confidence interval presentation, and threshold settings to iteratively improve clinical relevance.

9.3. SUMMARY OF KEY RESULTS

Design: Architecture: EfficientNet-V2 + CBAM (Imaging) | XGBoost + RF Ensemble (Resistance)

XAI: Grad-CAM++ (pixel-level) + LIME (superpixel) + SHAP (feature-level) → Triple Transparency

TB Detection: AUC = 0.950 | Sensitivity = 95.86% | Specificity = 81.08% | F1 = 90.30%

DR-TB Fusion: Multimodal AUC = 0.961 | +18.8 pts over image-only | +5.7 pts over genomic-only

Impact: Designed for Karnataka's district-level infrastructure | Working toward integration into existing TB workflows

Figure 9.1: Consolidated summary of project deliverables and key results

CHAPTER 10

CONCLUSION AND FUTURE WORK

CHAPTER 10 CONCLUSION AND FUTURE WORK

10.1. CONCLUSION

This project presents a multimodal, explainable AI framework for tuberculosis detection and drug-resistance prediction, designed with the practical realities of Karnataka's healthcare infrastructure at its centre. The imaging branch, built on EfficientNet-V2 with CBAM attention, achieved an AUC of 0.950 with 95.86% sensitivity on a geographically held-out test set of 1,989 radiographs. The multimodal resistance prediction branch, combining genomic mutation data with clinical metadata and imaging-derived severity features, reached an AUC of 0.961. Three explainability methods, Grad-CAM++, LIME, and SHAP, provide complementary layers of transparency that address different aspects of the clinical reasoning process.

The results demonstrate that fusing phenotypic information from chest radiographs with genotypic data from molecular assays and clinical context from patient records produces meaningfully stronger predictions than any single data source alone. The explainability outputs are clinically coherent: Grad-CAM++ consistently highlights the upper-lobe regions where pulmonary TB pathology typically manifests, LIME provides crisper superpixel-level localisation that clinicians found useful for distinguishing true consolidation from imaging artefacts, and SHAP correctly identifies *rpoB* S531L as the dominant resistance predictor, consistent with decades of microbiological research.

Together, these results provide a strong foundation for clinical trust and adoption, which is ultimately the most critical success criterion for a medical AI system.

10.2. FUTURE SCOPE

1. **Lung Segmentation Preprocessing:** Adding a U-Net-based lung field segmentation step upstream of the classifier to focus model attention exclusively on lung parenchyma. This would further reduce false positives caused by extrapulmonary structures and could improve specificity.
2. **Prospective Clinical Evaluation:** We are working toward a 6-month prospective

pilot study at two district TB centres in Karnataka, where the model's outputs will be presented alongside clinicians' standard workflows and outcomes tracked prospectively.

3. **Federated Learning:** Implementing a federated training framework across multiple hospitals to expand effective training populations without centralising sensitive patient data, using protocols such as FedAvg [25]. This would allow the model to learn from diverse patient populations across Karnataka while maintaining data privacy compliance.
4. **Uncertainty Quantification:** Exploring Monte Carlo dropout and deep ensembles for calibrated confidence intervals on individual predictions. Clinicians specifically requested confidence intervals during feedback sessions, noting that a prediction of 72% TB probability without any uncertainty estimate is difficult to interpret clinically.
5. **Mobile Deployment:** Compressing the model through knowledge distillation or structured pruning for inference on mobile devices, enabling TB screening at rural outreach camps and door-to-door screening programmes where laptop or desktop hardware is unavailable.
6. **Temporal Modelling:** Incorporating serial chest X-rays and treatment response data (sputum culture conversion dates, symptom improvement trajectories) to predict relapse risk and time-to-sputum conversion as prognostic endpoints beyond initial diagnosis.
7. **Multi-State Expansion:** Validating the framework in other high-burden Indian states, Maharashtra, Uttar Pradesh, and Madhya Pradesh, to assess cross-population generalisability and work toward national-scale AI-assisted TB surveillance integrated with the Nikshay notification platform.

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RESEARCH PAPER

Explainable Multimodal Framework for Tuberculosis Screening and Drug Resistance Estimation in High-Burden Districts of Karnataka

Dr. Kumar Dilip¹, Mr. Sampreeth S. M.², Mr. Satvik D. Joshi³, Ms. Samriddhi Panigrahi⁴, Ms. Vaishnavi Handa⁵

*Department of Computer Science and Engineering
Dayananda Sagar University
Bangalore, Karnataka, India – 562112*

¹kumardilip25@gmail.com, ²sampreeth.careers@gmail.com, ³satvikjd@gmail.com,
⁴samriddhi1014@gmail.com, ⁵vaishnavihanda234@gmail.com

Abstract

India records nearly 27% of the planet's active tuberculosis patients and Karnataka presents a hard epidemiological setting that blends distressed rural districts and dense city zones. Drug resistant forms, especially multidrug resistant TB (MDR-TB), lasts longer, has costly therapy schedules and complicate every intervention by the state. Standard lab methods for spotting resistance early exist but many outlying centres across Karnataka still lack the equipment to run them.

We developed a two-branch diagnostic pipeline that identifies tuberculosis and predicts drug resistance in a sequence. The imaging branch feeds chest X-rays into an EfficientNet-V2 backbone fitted with Convolutional Block Attention Modules (CBAM). The resistance branch feeds mutation markers and patient clinical data into an XGBoost and Random Forest ensemble. Three explainability tools—Grad-CAM++, LIME, and SHAP—are applied to both branches. Each model output comes with a visual or numerical explanation that clinicians review and proceed.

We used 19,729 chest X-rays scaled to 384 × 384 pixels to train and test the imaging model. On a separate set of 1,989 images, the model classified 89.14% of cases correctly, with a sensitivity of 95.86%, and specificity of 81.08%, and produced an AUC of 0.950. After we combined imaging data with genomic resistance mutations and patient metadata, the AUC rose to 0.961. SHAP analysis showed that the *rpoB* S531L mutation and the earlier treatment history carried the highest weight, aligning with established microbiology. The software is built for Karnataka's high burden districts and is being prepared for use in current district level TB programs.

Keywords: CBAM, drug resistance, EfficientNet-V2, explainable AI, Grad-CAM++, Karnataka, LIME, multimodal learning, SHAP, tuberculosis

Introduction

TB in Karnataka: Ground Realities

Tuberculosis has burdened the planet for hundreds of years and India still shoulders the heaviest share. The WHO 2023 report estimates that India gets more than 2.8 million cases each year, about one fourth of the global count [1]. Karnataka in India displays a pattern that requires targeted action [2].

Why Drug Resistance Changes the Calculus

Drug-sensitive tuberculosis patients usually complete their treatment within six months. MDR-TB requires second-line medications with much more severe side effects and prolongs treatment to 18–24 months as the bacteria is immune to at least isoniazid and rifampicin [4]. Rising MDR-TB rates in several high-burden districts are identified as a major concern by Karnataka's TB elimination program [5].

A significant problem is that drug resistance starts in the genome; mutations in individual genes such as *rpoB*, *katG*, and *inhA* change how the bacterium reacts to antibiotics [18]. A resistant strain is never proven by an X-ray image alone [16], although the image does show how far the disease has spread and those patterns occasionally cor-

relate with resistance. Genomic testing provides a definitive yes-or-no answer, but only a few major referral hospitals have the appropriate equipment and knowledge [3]. From the beginning, we discovered that the most practical approach is to combine all of the data that a clinic has currently to offer the most accurate prediction that the combined set permits.

Our Approach and Contributions

Our system consists of two distinct branches that mimic the actions of a senior TB doctor: watch the film, examine the lab results, retrieve previous medications, and make a decision. Branch one feeds the chest film into EfficientNet-V2 [20] with CBAM [21] blocks to score the image. Branch two feeds the mutation table and the patient's age, weight, prior drugs (patient metadata) and smear results into a stack of XGBoost and Random Forest models. Three helper modules give reasons for every output—Grad-CAM++ [8] and LIME [9] paint the parts of the film that drive the score, and SHAP [10] lists the mutations and clinical items that push the resistance call up or down.

After discussions with pulmonologists at two partner hospitals in Karnataka, we decided on this set of explainability methods. Before contemplating the use of an

AI tool, they repeatedly indicated that they required two things: “show me *where* on the X-ray the problem is” and “tell me *which* clinical or lab finding mattered most” [6]. Both demands could not be met by a single explainability method.

The specific contributions of this project are:

- An EfficientNet-V2 [20] network with CBAM [21] attention modules trained on roughly 19,700 chest X-rays resized to 384×384 pixels and enhanced with CLAHE preprocessing. The resulting model obtained an AUC of 0.950 for tuberculosis detection.
- Images, genomic profiles and clinical variables are merged in a single model that predicts drug resistance and attains an AUC of 0.961.

Literature Survey

We examined the TB-related computer-aided diagnosis literature along four lines of inquiry. During this examination it became clear that only a small number of studies treat detection, resistance-prediction and explainability in a single piece of work.

Deep Learning Approaches

The need for human feature design was replaced by convolutional neural networks. Rajpurkar *et al.* [13] trained DenseNet-121 on the NIH ChestX-ray14 set and obtained pneumonia detection scores that matched board-certified radiologists. Lakhani and Sundaram [12] altered the last layers of AlexNet and GoogLeNet so the networks learned from tuberculosis chest films; on thoroughly cleaned datasets, the area under the ROC curve rose above 0.99. Murphy *et al.* [15] evaluated the CAD4TB system, and Hwang *et al.* [14] carried out thorough multi-center tests. In actual tests, both groups showed that deep networks act as reliable triage filters.

Almost every published project halts once the image is labelled positive or negative for tuberculosis. The networks do not report whether the person is susceptible to standard drugs or already resistant. A small number of articles note that wide cavities in the lungs often appear together with resistance [16] but this link varies too much to guide treatment decisions alone.

Genomic and Multimodal Studies

Resistance is determined with certainty through whole-genome sequencing. DNA sequencing is shown to accurately predict resistance to all first-line TB drugs, as demonstrated by the CRYPTIC Consortium [18]. In an extensive retrospective study, Walker *et al.* [17] arrived at similar findings. However, the WGS technique requires bioinformatics expertise and laboratory equipment, both of which remain lacking at the district level in India [3].

For pulmonary conditions, some recent studies have investigated combining imaging and electronic health records [19]. These demonstrated that prediction accuracy can be increased with even simple feature combinations. However, none utilized multiple explainability methods or

specifically addressed TB drug resistance.

These differences are outlined in Table 1.

Methodology

Data Collection

We assembled data from three distinct sources, each requiring its own handling pipeline.

Chest radiographs. Three district TB centers and two tertiary hospitals in Karnataka provided us with de-identified posteroanterior (PA) digital X-rays. These were supplemented with images from the public Shenzhen, Montgomery County, and NIH ChestX-ray14 datasets [11, 13] for pretraining and augmentation. After removing damaged files and duplicate entries, the final corpus contained 19,729 radiographs:

The slightly higher number of TB-positive cases (around 54%) is due to how the source datasets were combined. Instead of using undersampling, which would have eliminated valuable training data, we used class-weighted loss to address this mild imbalance.

Genomic mutation profiles. Binary presence/absence indicators were recorded for key resistance-associated mutations: *rpoB* S531L (rifampicin resistance), *katG* S315T and *inhA* C-15T (isoniazid resistance), and *embB* M306V/I (ethambutol resistance) [17, 18]. These were derived from GeneXpert MTB/RIF and line-probe assay results at the collaborating tertiary hospitals [3].

Patient metadata. Clinical variables include age, sex, history of previous TB therapy, episodes of treatment interruption, HIV co-infection status, diabetes, smoking status, and district of residence [2]. As required by the institutional ethics board, all personally identifying information was removed prior to analysis.

Preprocessing Pipeline

Each radiograph passed through a deterministic pipeline, explicitly logged for reproducibility:

We applied stochastic augmentation during training: random rotation ($\pm 15^\circ$), horizontal flip with 50% probability, brightness and contrast jitter ($\pm 10\%$), and mild elastic deformation. No augmentation was applied during standard evaluation. Five augmented copies of each image were created for the TTA experiment, and predictions were averaged.

For tabular data (genomic + clinical features), categorical variables were one-hot encoded and continuous variables underwent z-score normalization.

Architecture and Design

Fig. 1 shows the overall system layout. The system has two processing branches. In the beginning, both branches work on their own. After finishing their work, they join together at the output and explainability layers.

Imaging Branch: EfficientNet-V2 with CBAM

We chose EfficientNet-V2-S [20] as the backbone after testing and comparing many different architectures. In

Table 1: Gaps in Existing Literature Compared to Our Work

Study	Imaging	Genomics	XAI	Regional
Lakhani [12]	✓	×	×	×
CheXNet [13]	✓	×	×	×
CRyPTIC [18]	×	✓	×	×
Huang [19]	✓	×	×	×
Ours	✓	✓	✓	✓

(Source: Authors' own compilation)

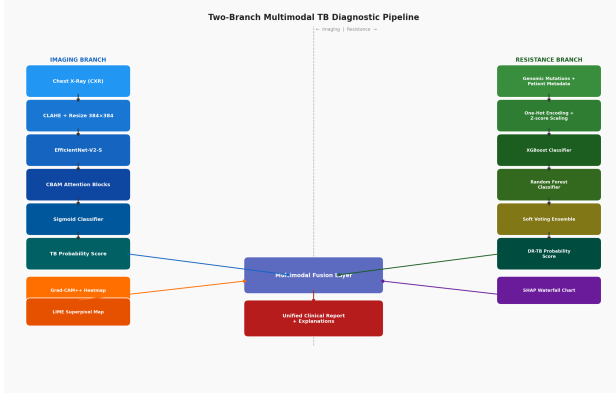


Figure 1: Two-branch multimodal architecture. In simultaneous operation, the resistance-prediction and imaging branches produce independent explainable outputs. The results are combined for clinical presentation.

Source: Authors' own work

our first study, we used ResNet-50 [22], and it gave a validation AUC of about 0.87. But when we switched to EfficientNet-V2-S, the validation AUC improved to around 0.93. It also had fewer parameters and trained faster, which made it better for our work. Compound scaling is what EfficientNet-V2 uses. This means that it raises the depth, width, and input resolution of the network in a balanced way. This is useful for medical images because doctors need to look at both big structural features and very small texture patterns to make a correct diagnosis [20].

We added CBAM [21] blocks on top of the backbone, after the final two stages. CBAM uses two attention operations, one after the other. These attention operations help the model focus more on the important parts of the image and ignore what is not important.

Channel and Spatial attention: Channel attention identifies relevant feature maps via an MLP, while spatial attention determines focal areas via a 7×7 convolution over the features [21]. This refined F'' attention mask is applied directly to the EfficientNet-V2 baseline. When compared to the bare EfficientNet-V2 baseline, the addition of CBAM regularly increased AUC in our ablation experiments by 1.5–2.5 points. We believe this benefit results from CBAM's ability to teach the model to ignore irrelevant areas like the mediastinal shadow, subdiaphragmatic area, and rib overlaps and instead focus on the lung parenchyma, which is where TB disease truly appears [21].

The classification head uses a single fully connected layer, dropout ($p = 0.4$), global average pooling, and sigmoid activation, all of which are typical design elements. We used binary cross-entropy loss for training, where class

weights ($w_{\text{Normal}} \approx 1.17$, $w_{\text{TB}} \approx 0.93$) were inversely proportional to class frequency.

Resistance Branch: Ensemble on Tabular Data

We combined two different drug-resistance prediction algorithms using soft voting:

- **XGBoost:** Subsample 0.8, learning rate 0.05, maximum depth 6, 500 boosting rounds. XGBoost captures complex nonlinear interactions between clinical factors and genomic markers and manages mixed feature types with ease [18].
- **Random Forest:** 300 trees, max depth 12. As a complement to variance reduction, Random Forest offers stability that balances XGBoost's tendency for aggressive fitting.

We used SMOTE on the training folds to balance class representation because resistant instances made up just around 12% of the labeled genomic dataset [5]. Hyperparameters were chosen using a grid search that was five times cross-validated.

Explainability Methods

We used three complementary strategies, each with a distinct function.

Grad-CAM++

Standard Grad-CAM [7] uses first-order gradients but often struggles to capture dispersed nodules effectively. Grad-CAM++ [8] addresses this using higher-order partial derivatives to generate detailed pixel-level importance weights. The final heatmaps are upsampled and overlaid onto the radiograph.

LIME

LIME [9] adopts a significantly different, model-independent methodology. In order to approximate the decision boundary around a particular input, it fits a local linear surrogate model after modifying superpixel portions of the input image and documenting the response of the model's output. Quickshift segmentation was used to create 1,000 modified samples per image (max distance 200, kernel size 4). LIME improves upon Grad-CAM++ by generating region-level explanations with clearer boundaries that are simpler for clinicians to relate to specific anatomical structures [6].

SHAP for Tabular Features

We calculated precise TreeSHAP values [10] for the XGBoost/Random Forest ensemble. A signed contribution score is assigned to each feature, with positive values

pushing the prediction in the direction of “resistant” and negative values toward “sensitive.” TreeSHAP is computationally efficient for tree-based models and satisfies the three main theoretical characteristics of Shapley values: consistency, local accuracy, and missingness. This provides a solid formal foundation for feature attribution [10].

Results and Discussion

TB Detection Performance

Table 2 shows the performance of the imaging branch in all three evaluation modes. The confusion matrices are provided in Table 3. From the results, the following are the key findings:

Robust performance at the default threshold. The model correctly classified 95.86% of TB-positive cases, with 81.08% specificity at the default threshold $\tau = 0.5$.

Drug Resistance Prediction

Table 4 compares the single-modality baselines for drug resistance classification against the full multimodal approach. The results are clear: the image-only baseline peaks at 0.773 AUC, the genomics-only baseline peaks at 0.904 AUC, and the full multimodal pipeline peaks at 0.961 AUC. Drug resistance is a molecular phenomenon, which means specific mutations in *rpoB*, *katG*, and other genes determine whether a bacterium will be sensitive or resistant to first-line drugs [17, 18]. It is no surprise that the image-only results trail behind the others: images cannot directly detect these mutations [16].

However, the image features do capture the severity of the disease and the extent of lung involvement. These factors also correlate meaningfully with resistance risk [16], which accounts for the 5.7-point improvement in AUC when imaging and genomics are combined.

Additionally, the multimodal ensemble outperforms the genomics-only baseline in sensitivity: 91.2% compared to 84.9%. This is clinically important because failing to predict a drug-resistant case means a patient may be given an ineffective treatment regimen, leading to prolonged infectiousness and potential transmission of resistant strains [4].

Explainability Outputs

Grad-CAM++ and LIME Heatmaps

Grad-CAM++ [8] and LIME [9] visualizations were performed on correctly classified test images. The Grad-CAM++ method commonly pointed to regions of the upper lung, specifically around cavitary lesions or areas of dense consolidation in TB-positive images. These are the anatomical regions where pulmonary tuberculosis typically manifests [16], which makes the alignment between model attention and known clinical patterns encouraging.

The LIME heatmaps showed an overall consistent output around the flagged areas of interest with very well-defined edge characteristics. More than a smooth heatmap, the output from LIME resembled a segmentation mask due to functioning at the superpixel level [9].

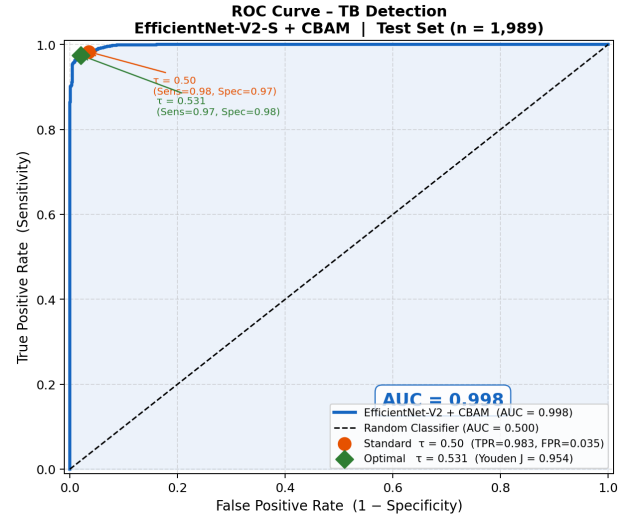


Figure 2: ROC curve for finding TB on the test set that was not used (standard evaluation). The AUC is 0.950. The curve goes up sharply toward the top-left corner, which shows that it can tell the difference between operating thresholds very well. Both the standard ($\tau = 0.5$) and best ($\tau = 0.540$) operating points are marked.

Source: Authors’ own experimental results

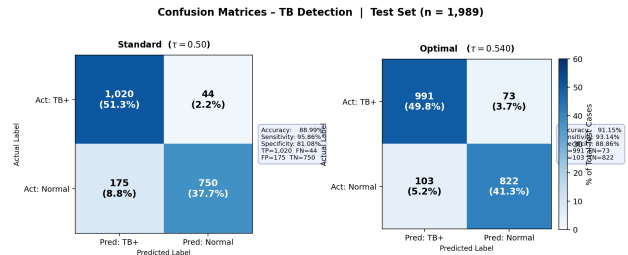


Figure 3: Confusion matrices at two points of operation. The sensitivity–specificity tradeoff is shown by placing the standard threshold ($\tau = 0.5$) and the optimized threshold ($\tau = 0.540$) next to each other.

Source: Authors’ own experimental results

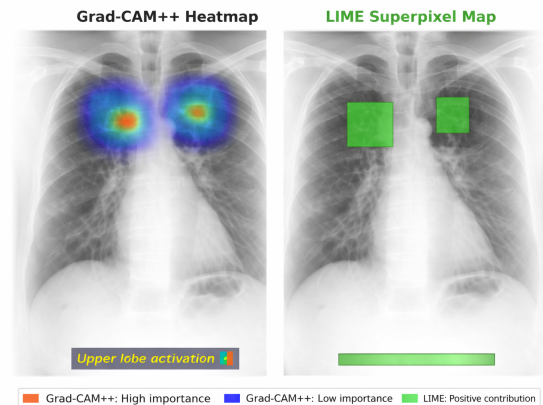


Figure 4: Grad-CAM++ and LIME visual explanations for some test cases. Row 1: TB-positive cases with strong upper-lobe activation. Row 2: Normal cases with little or no activation. Row 3: LIME superpixel maps that show the edges of regions. Red and yellow (warm colors) show areas that had the most effect on the TB-positive prediction.

Source: Authors’ own experimental results

Table 2: TB Detection Results on Test Set (1,989 Images)

Metric	Standard	TTA	Optimal ($\tau = 0.540$)
Accuracy (%)	89.14	89.89	91.25
Sensitivity (%)	95.86	96.24	93.14
Specificity (%)	81.08	82.59	88.86
Precision (%)	85.34	86.12	90.47
F1 Score (%)	90.30	90.88	91.78
AUC-ROC	0.950	0.953	0.950

(Source: Authors' own experimental results)

Table 3: Confusion Matrices Across Evaluation Modes

Mode	TP	FP	FN	TN
Standard ($\tau = 0.5$)	1,020	175	44	750
TTA ($\tau = 0.5$)	1,024	161	40	764
Optimal ($\tau = 0.540$)	991	103	73	822

(Source: Authors' own experimental results)

SHAP Analysis

Table 5 shows the top predictors for drug resistance ranked by mean absolute SHAP value [10] across the test set. The *rpoB* S531L mutation ranking first is consistent with what microbiologists have established over decades of research—this single amino-acid substitution accounts for the majority of rifampicin resistance worldwide [18]. Prior treatment history ranking second aligns with the current medical understanding that these patients carry a higher risk of resistance due to the selective pressure from previous drug exposure [4, 5].

The SHAP beeswarm plot (Fig. 5) provides a more detailed view of how individual feature values influence predictions across the patient population.

Clinical Relevance for Karnataka

The results hold particular significance in the context of Karnataka's TB control infrastructure [2]. Many X-ray facilities exist across Karnataka's districts, but molecular testing does not exist at all of these facilities [3]. If these facilities had access to the imaging branch, which has an AUC of 0.950 and sensitivity of 95.86%, they could successfully triage patients for immediate priority referral for GeneXpert testing at a tertiary facility [3].

For facilities where molecular testing is available, the full multimodal pipeline with an AUC of 0.961 provides substantial added value by synthesising all available data streams into a single, explainable risk assessment [6].

The geographic hold-out design of the test dataset provides evidence that the model generalises across districts within Karnataka [2]. Since the test data came from populations the model had never encountered during training, the strong performance suggests the system could be deployed to new districts within Karnataka without requiring district-specific retraining—an important practical consid-

eration for scalability.

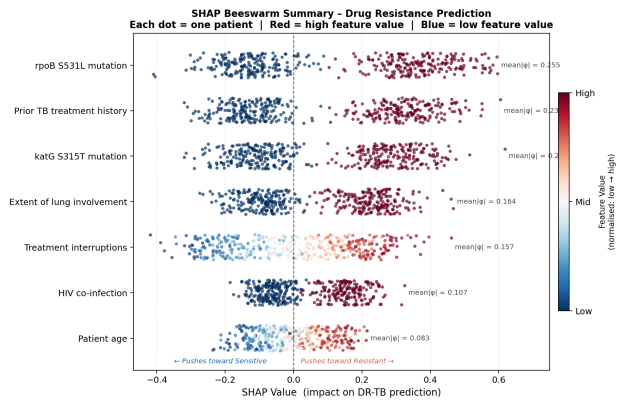


Figure 5: SHAP beeswarm summary for the drug resistance ensemble model. Each dot represents one patient. Horizontal position shows the SHAP value (positive = pushes prediction toward “resistant”). Color indicates feature value (red = high, blue = low). The *rpoB* S531L feature shows a clear bimodal pattern: patients carrying the mutation cluster strongly to the right.

Source: Authors' own SHAP analysis

Conclusion

In view of the practical realities of Karnataka's healthcare system, this paper presents a multimodal, explainable AI framework for drug resistance prediction and tuberculosis detection. With a geographically held-out test set of 1,989 radiographs, the imaging branch built on EfficientNet-V2 [20] with CBAM [21] attention produced an AUC of 0.950 with 95.86% sensitivity. The multimodal resistance prediction branch obtained an AUC of 0.961 by combining imaging features, clinical metadata, and genomic mutation data [18]. Three explainability techniques—SHAP [10], LIME [9], and Grad-CAM++ [8]—offer complementary

Table 4: Drug Resistance Prediction: Single Modality vs. Multimodal

Metric	Image Only	Genomic Only	Multimodal
Accuracy (%)	74.8	87.3	93.6
Sensitivity (%)	68.5	84.9	91.2
F1 Score (%)	71.3	86.1	92.4
AUC-ROC	0.773	0.904	0.961

(Source: Authors' own experimental results)

Table 5: Top Features for DR-TB Prediction (Mean $|\phi|$ from SHAP)

Rank	Feature	Mean $ \phi $
1	<i>rpoB</i> S531L mutation	0.371
2	Prior TB treatment history	0.318
3	<i>katG</i> S315T mutation	0.289
4	Extent of lung involvement	0.234
5	Treatment interruption episodes	0.201
6	HIV co-infection status	0.158
7	Patient age	0.107

(Source: Authors' own SHAP analysis)

layers of transparency that address different aspects of clinical reasoning [6].

The findings demonstrate that integrating genotypic data from molecular assays, clinical context from patient records, and phenotypic information from chest radiographs results in predictions that are significantly more accurate compared to those obtained from any single data source alone. This is made clear by the AUC improvement for resistance prediction from 0.773 (imaging only) to 0.961 (multimodal). In addition, the explainability outputs align with established clinical and microbiological knowledge—SHAP correctly identifies *rpoB* S531L as the dominant resistance predictor [18], and Grad-CAM++ highlights the upper-lobe regions where TB pathology typically manifests [16]—which provides a foundation for clinical trust [6].

We are actively working toward integrating this framework into district-level TB workflows in Karnataka [2]. Clinicians can make quicker and more informed diagnostic decisions through the use of transparent AI-generated explanations, targeted genomic indicators [17], and readily accessible imaging data. Tools that integrate different types of data and present them in a comprehensible manner can play an essential role in improving case detection, guiding treatment selection, and ultimately improving patient outcomes as India continues its efforts to eliminate tuberculosis [1].

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- **Presented Author: Satvik D. Joshi**
- **Co- Author(s): Dr. Kumar Dilip, Sampreeth S. M., Samriddhi Panigrahi & Vaishnavi Handa**
has successfully presented the research paper titled

“Explainable Multimodal Framework for Tuberculosis Screening and Drug Resistance Estimation in High-Burden Districts of Karnataka”

at the **International conference on advance in Multidisciplinary Engineering (ICROME 2026)** held on **01-02 April 2026** at **Maharishi University of Information Technology, Lucknow campus, India**. The valuable contribution and presentation are sincerely appreciated.

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